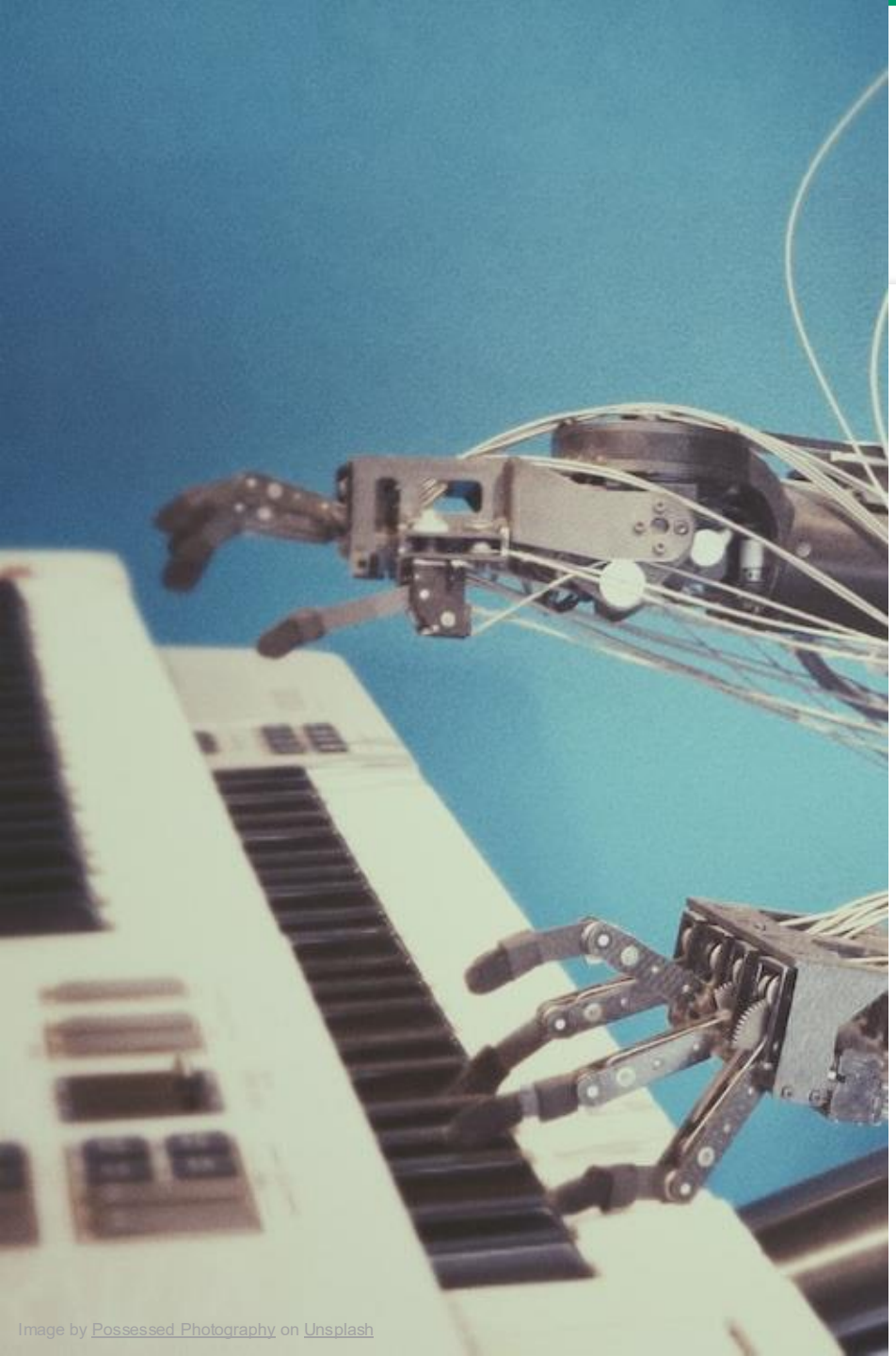


Applied Artificial Intelligence
01 - Foundations

Univ.-Prof. Dr-Ing. habil. Niklas Kühl
www.niklas.xyz

University of Bayreuth

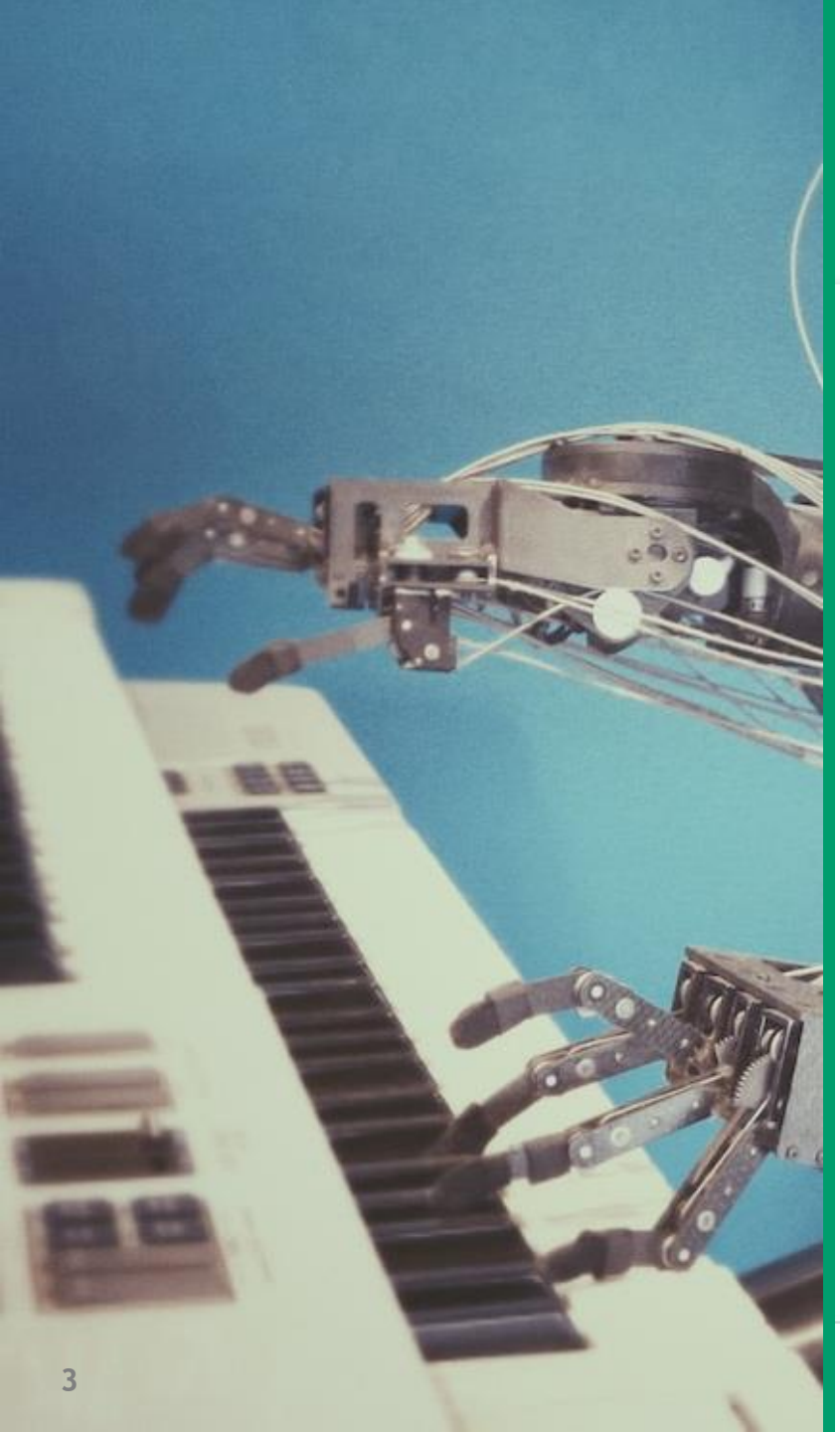
TUM School of Management

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Objectives of this lecture

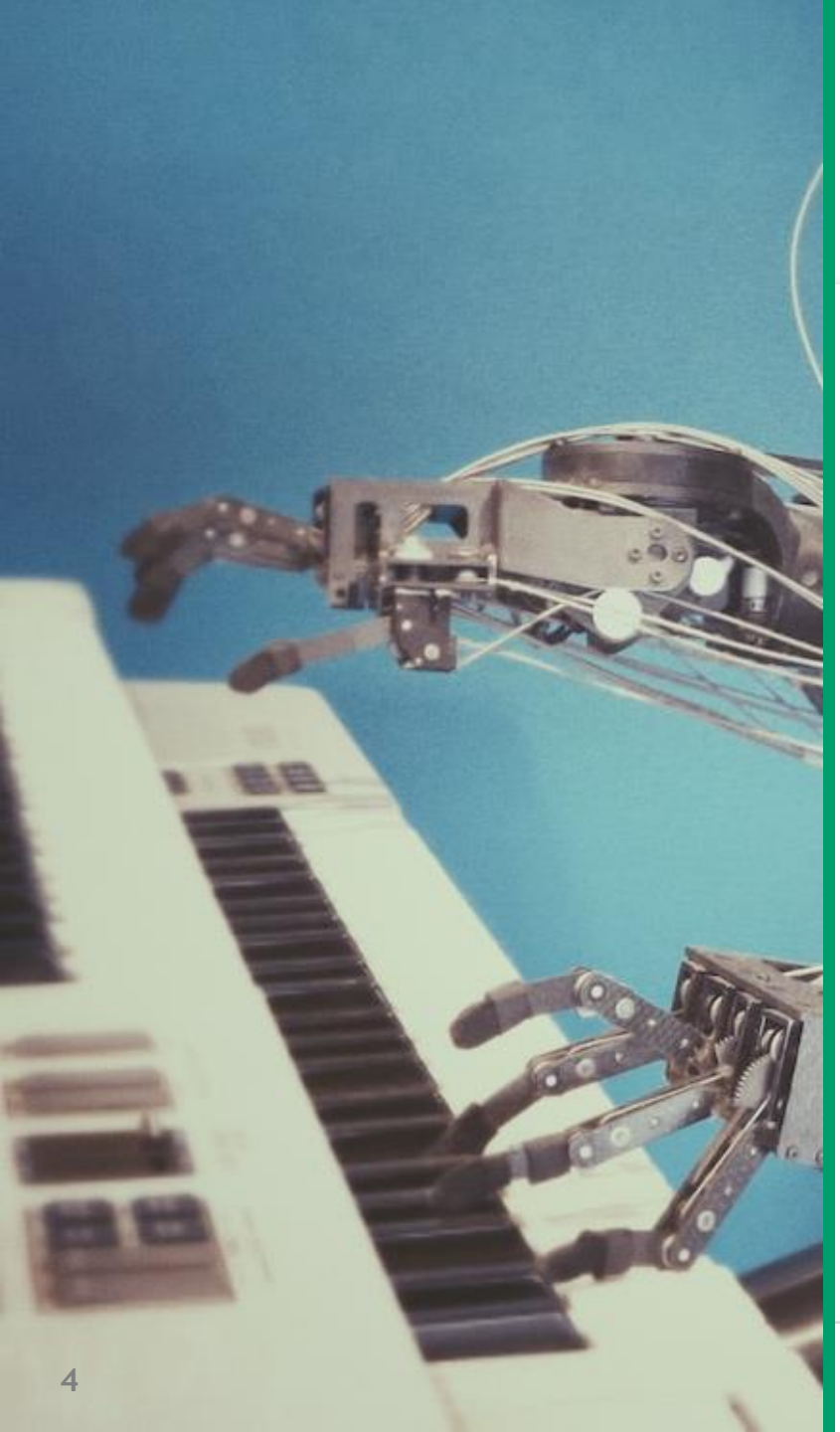
- Become familiar with the terminology, in particular Artificial Intelligence, Machine Learning and Systems Thinking
- Gain a first understanding of the AI lifecycle

Prof. Dr. Niklas Kühl



1. Organizational
2. Artificial Intelligence & Machine Learning
3. Systems Thinking
4. The AI Lifecycle

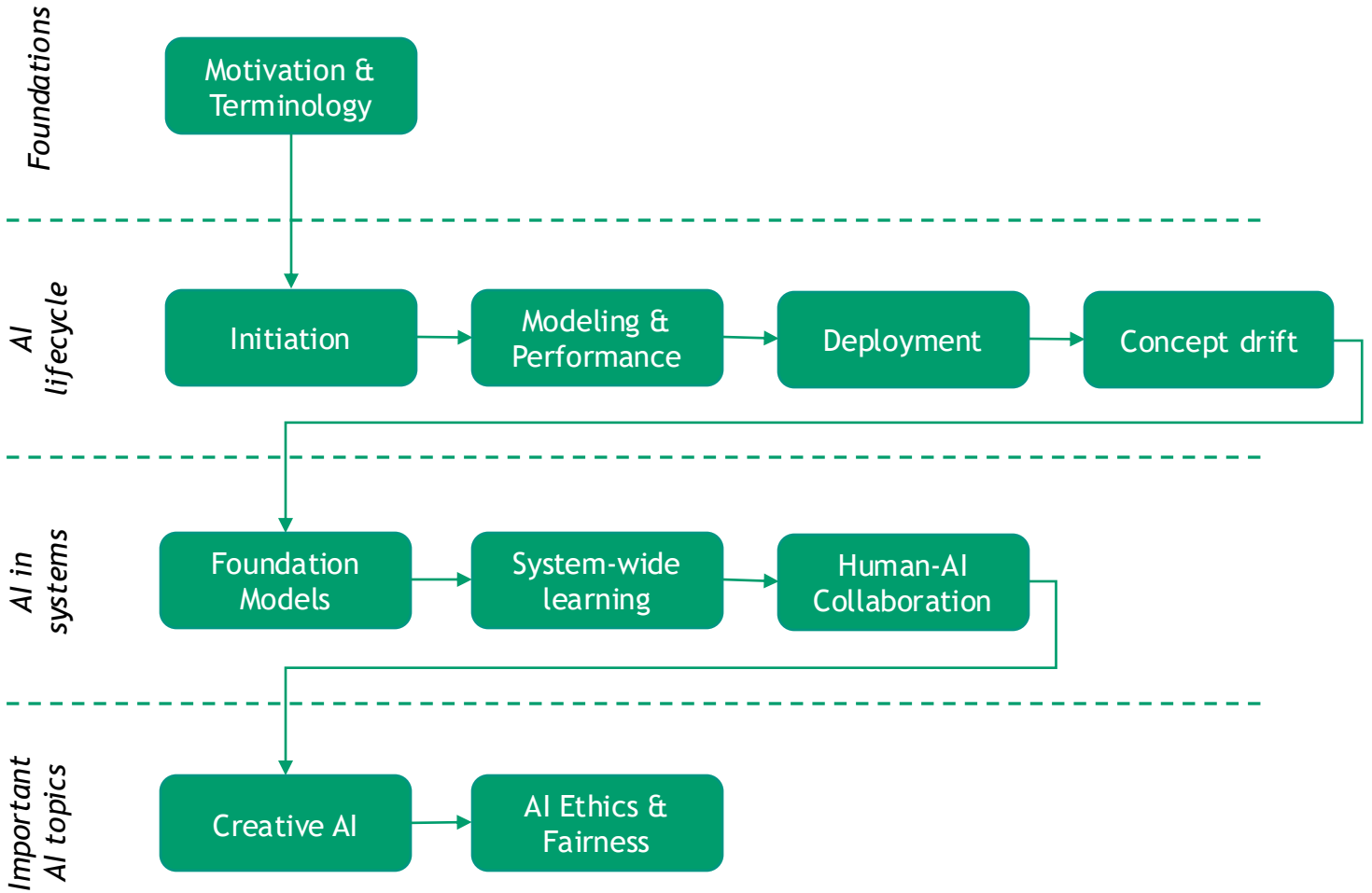
Prof. Dr. Niklas Kühl

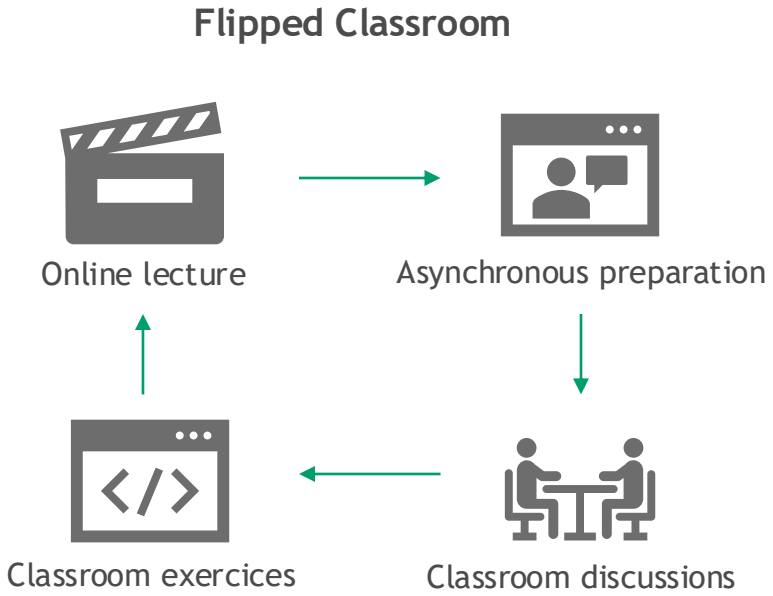


1. **Organizational**
2. Artificial Intelligence & Machine Learning
3. Systems Thinking
4. The AI Lifecycle

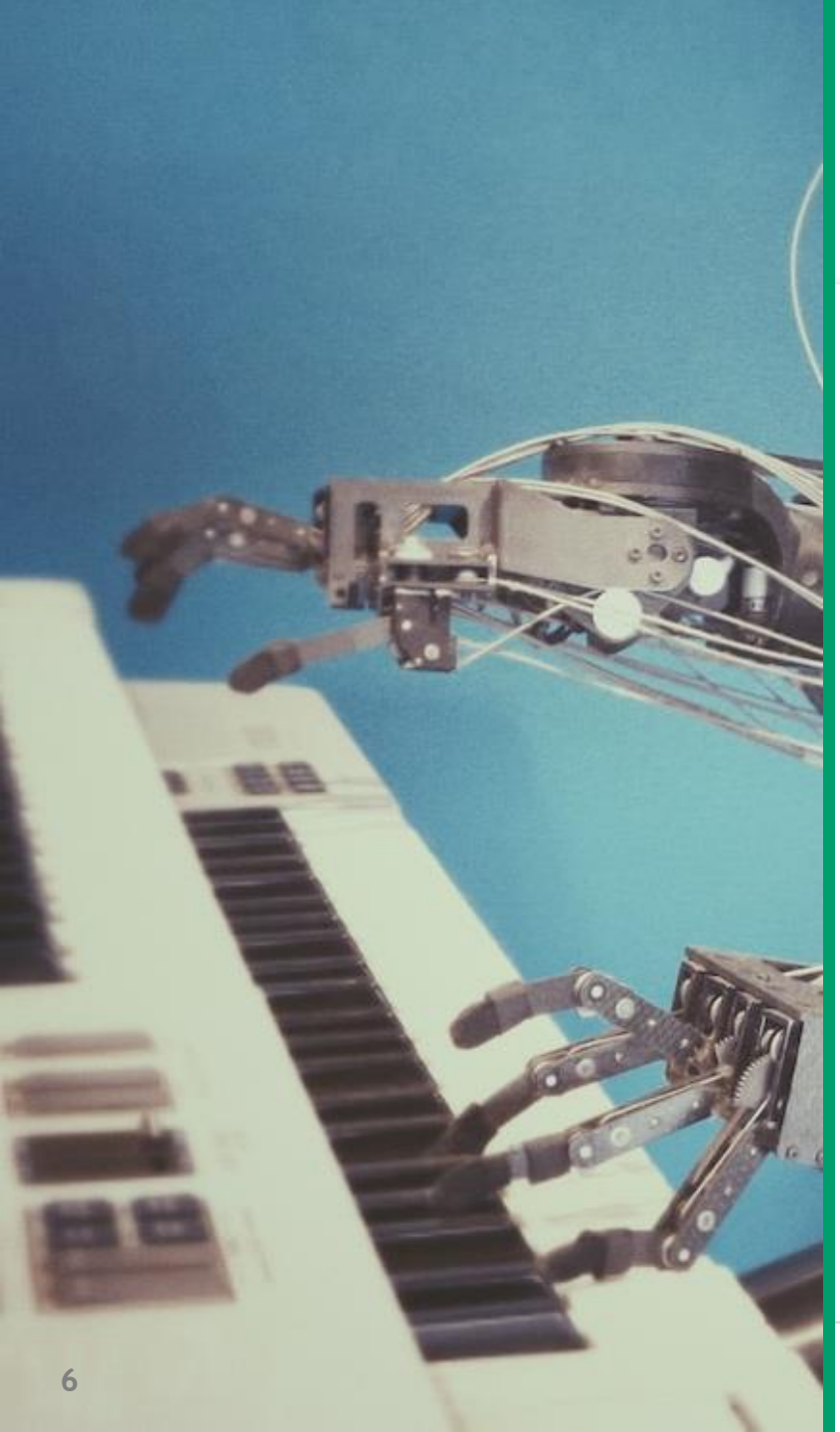
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Organizational
The story of the lecture





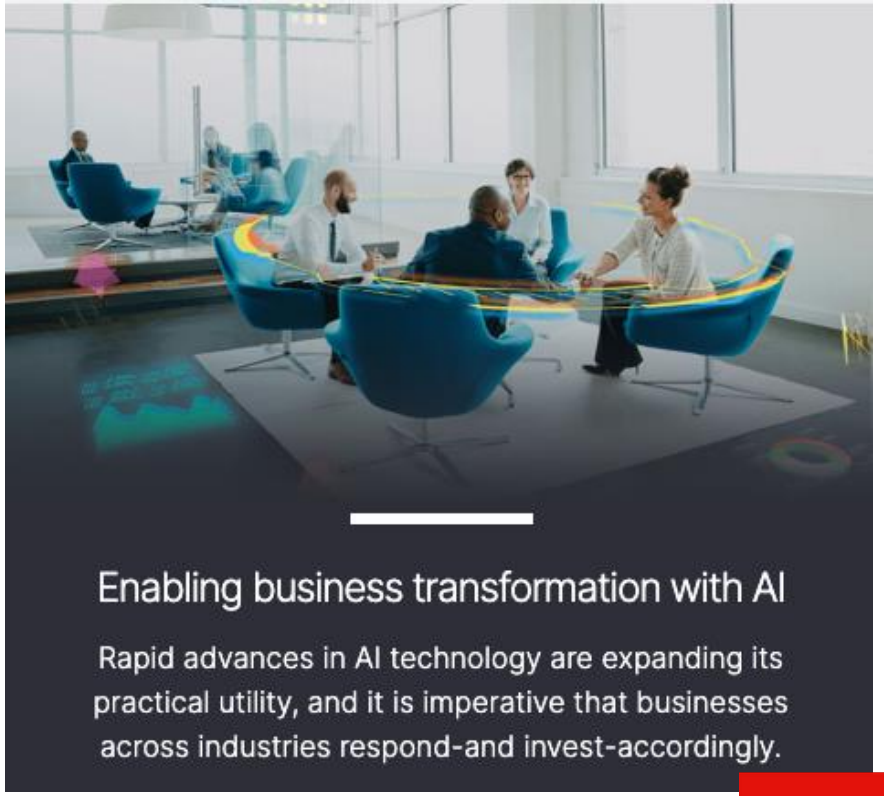
Prof. Dr. Niklas Kühl



1. Organizational
2. **Artificial Intelligence & Machine Learning**
3. Systems Thinking
4. The AI Lifecycle

Prof. Dr. Niklas Kühl

Artificial Intelligence & Machine Learning
Why AI? A business perspective.



[1]

February 12th 2024

The Economist

Artificial intelligence is driving real business innovation

How organizations can build and scale game-changing generative AI services



The Washington Post

[2]

June 14th 2024

https://impact.economist.com/projects/facing-the-future-with-ai/enabling-business-transformation-with-ai/ [1]
https://www.washingtonpost.com/creativegroup/aws/artificial-intelligence-is-driving-real-business-innovation/ [2]

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Artificial Intelligence & Machine Learning

Why AI? A research perspective.



www.nature.com/jhg

REVIEW ARTICLE OPEN

Advances in AI and machine learning for predictive medicine

Alok Sharma [1,2,3], Artem Lysenko [1,2], Shangru Jia [4], Keith A. Boroevich [2] and Tatsuhiko Tsunoda [1,2,4]

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Health Sciences Review 10 (2024) 100150

Contents lists available at ScienceDirect

Health Sciences Review

journal homepage: www.elsevier.com/locate/hsr

Systematic Review / Meta-analysis

Towards integration of artificial intelligence into medical devices as a real-time recommender system for personalised healthcare: State-of-the-art and future prospects

Talha Iqbal [a,b,*], Mehedi Masud [a], Bilal Amin [a,c], Conor Feely [d], Mary Faherty [d], Tim Jones [d], Michelle Tierney [d], Atif Shahzad [a,e], Patricia Vazquez [a]

[a] Smart Sensor Lab, Lambe Institute of Translational Research, College of Medicine, Nursing Health Sciences, University of Galway, Galway H91 TK33, Ireland
[b] Insight SFI Centre for Data Analytics, Galway H91 TK33, Ireland
[c] Electrical and Electronic Engineering, University of Galway, Galway H91 TK33, Ireland
[d] Symphysis Medical, ATU Innovation Hubs H91 DCH9, Ireland
[e] Centre for Systems Modelling and Quantitative Biomedicine (SMQB), University of Birmingham B15 2TT, United Kingdom

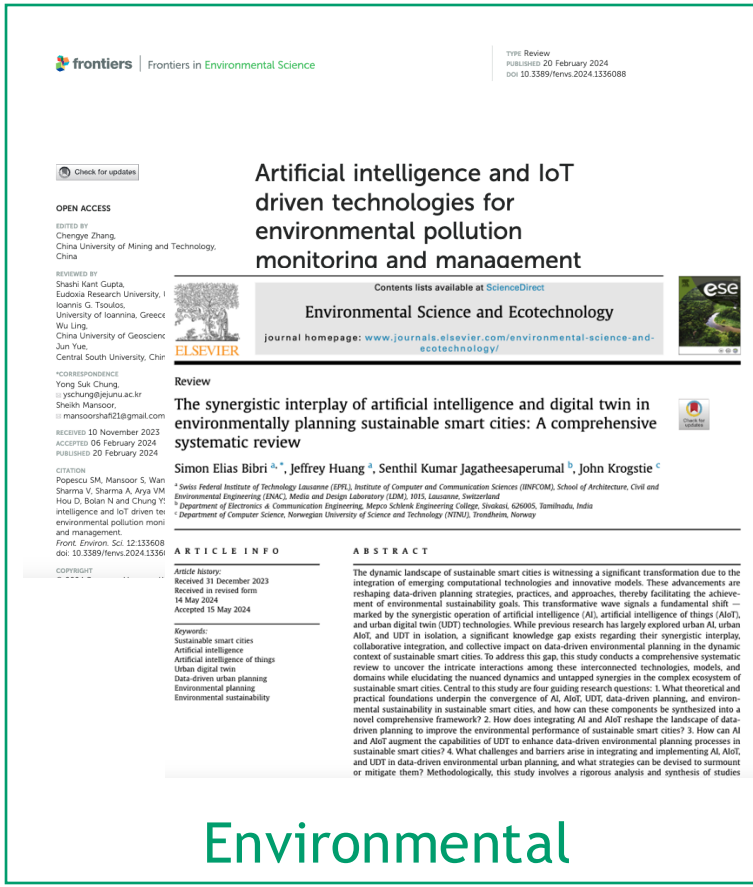
ARTICLE INFO

Keywords:
Artificial intelligence
Recommender systems
Personalized healthcare
Performance validation

ABSTRACT

In the era of big data, artificial intelligence (AI) algorithms have the potential to revolutionize healthcare by improving patient outcomes and reducing healthcare costs. AI algorithms have frequently been used in health care for predictive modelling, image analysis and drug discovery. Moreover, as a recommender system, these algorithms have shown promising impacts on personalized healthcare provision. A recommender system learns the behaviour of the user and predicts their current preferences (recommends) based on their previous preferences. Implementing AI as a recommender system improves this prediction accuracy and solves cold start and data sparsity problems. However, most of the methods and algorithms are tested in a simulated setting which cannot recapitulate the influencing factors of the real world. This review article systematically reviews prevailing methodologies in recommender systems and discusses the AI algorithms as recommender systems specifically in the field of healthcare. It also provides discussion around the most cutting-edge academic and practical contributions present in the literature, identifies performance evaluation matrices, challenges in the implementation of AI as a recommender system, and acceptance of AI-based recommender systems by clinicians. The findings of this article direct researchers and professionals to comprehend currently developed recommender systems and the future of medical devices integrated with real-time recommender systems for personalized healthcare.

Medical



frontiers | Frontiers in Environmental Science

TYPE Review
PUBLISHED 20 February 2024
DOI 10.3389/fenvs.2024.1336088

Artificial intelligence and IoT driven technologies for environmental pollution monitoring and management

OPEN ACCESS

Contents lists available at ScienceDirect

Environmental Science and Ecotechnology

journal homepage: www.journals.elsevier.com/environmental-science-and-ecotechnology/

Review

The synergistic interplay of artificial intelligence and digital twin in environmentally planning sustainable smart cities: A comprehensive systematic review

Simon Elias Bibri [a,*], Jeffrey Huang [a], Senthil Kumar Jagatheesaperumal [b], John Krogstie [c]

[a] Swiss Federal Institute of Technology Lausanne (EPFL), Institute of Computer and Communication Sciences (IINFCOM), School of Architecture, Civil and Environmental Engineering (ENAC), Media and Design Laboratory (LDM), 1015, Lausanne, Switzerland
[b] Department of Electronics & Communication Engineering, Mepco Schlenk Engineering College, Sivakasi, 626005, Tamilnadu, India
[c] Department of Computer Science, Norwegian University of Science and Technology (NTNU), Trondheim, Norway

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Sustainable smart cities
Artificial intelligence
Artificial intelligence of things
Urban digital twin
Data-driven urban planning
Environmental planning
Environmental sustainability

ABSTRACT

The dynamic landscape of sustainable smart cities is witnessing a significant transformation due to the integration of emerging computational technologies and innovative models. These advancements are reshaping data-driven planning strategies, practices, and approaches, thereby facilitating the achievement of environmental sustainability goals. This transformative wave signals a fundamental shift — marked by the synergistic operation of artificial intelligence (AI), artificial intelligence of things (AIoT), and urban digital twin (UDT) technologies. While previous research has largely explored urban AI, urban AIoT, and UDT in isolation, a significant knowledge gap exists regarding their synergistic interplay, collaborative integration, and collective impact on data-driven environmental planning in the dynamic context of sustainable smart cities. To address this gap, this study conducts a comprehensive systematic review to uncover the intricate interactions among these interconnected technologies, models, and domains while elucidating the nuanced dynamics and untapped synergies in the complex ecosystem of sustainable smart cities. Central to this study are four guiding research questions: 1. What theoretical and practical foundations underpin the convergence of AI, AIoT, UDT, data-driven planning, and environmental sustainability in sustainable smart cities, and how can these components be synthesized into a novel comprehensive framework? 2. How does integrating AI and AIoT reshape the landscape of data-driven planning to improve the environmental performance of sustainable smart cities? 3. How can AI and AIoT augment the capabilities of UDT to enhance data-driven environmental planning processes in sustainable smart cities? 4. What challenges and barriers arise in integrating and implementing AI, AIoT, and UDT in data-driven environmental urban planning, and what strategies can be devised to surmount or mitigate them? Methodologically, this study involves a rigorous analysis and synthesis of studies

Environmental



www.nature.com/scientificreports

scientific reports

OPEN

Opportunities and challenges of integrating artificial intelligence

Information Development
OnlineFirst
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https://doi.org/10.1177/02666669231200628

Sage Journals

Original Manuscript

Artificial intelligence in developing countries: The impact of generative artificial intelligence (AI) technologies for development

Nishith Reddy Mannuru, Sakib Shahriar, Zoë A Teel, Ting Wang, Brady D Lund, Solomon Tijani, Chalermchai Oak Pohboon, Daniel Agbaji, Joy Alhassan, JaKLyn Galley, Raana Kousari, Lydia Ogbadu-Oladapo, Shubham Kumar Saurav, Aishwarya Srivastava, Sai Priya Tummuru, Sravya Uppala, and Praveenkumar Vaidya

Abstract

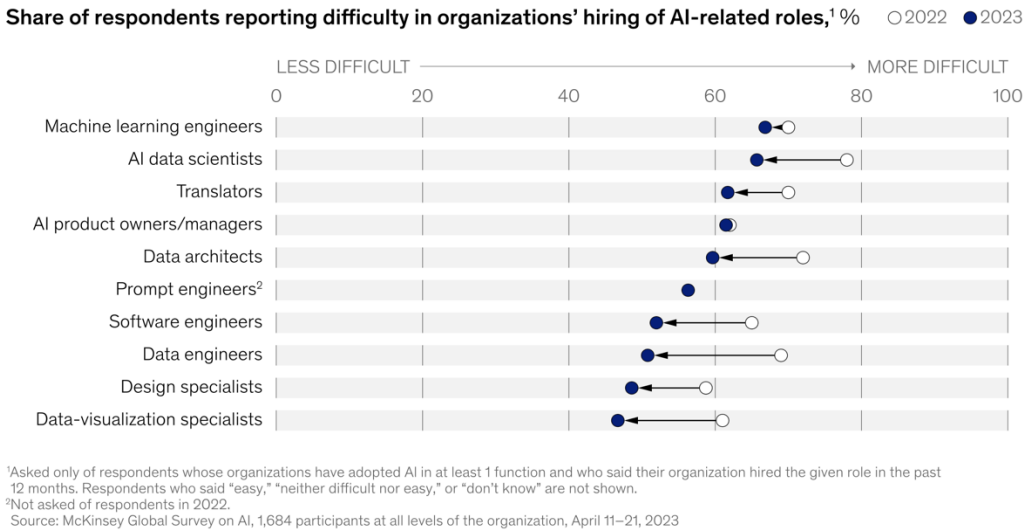
This paper explores the potential impact of Generative Artificial Intelligence (Generative AI) on developing countries, considering both positive and negative effects across various domains of information, culture, and industry. Generative Artificial Intelligence refers to artificial intelligence (AI) systems that generate content, such as text, audio, or video, aiming to produce novel and creative outputs based on training data. Compared to conversational artificial intelligence, generative artificial intelligence systems have the unique capability of not only providing replies but also generating the content of those responses. Recent advancements in Artificial Intelligence during the Fourth Industrial Revolution, exemplified by tools like ChatGPT, have gained popularity and reshaped content production and creation. However, the benefits of generative artificial intelligence are not equally accessible to all, especially in developing countries, where limited access to cutting-edge technologies and inadequate infrastructure pose challenges. This paper seeks to understand the potential impact of generative AI technologies on developing countries, considering economic growth, access to technology, and the potential paradigm shift in education, healthcare, and the environment. The findings emphasize the importance of providing the necessary support and infrastructure to ensure that generative AI contributes to inclusive development rather than deepening existing inequalities. The study highlights the significance of integrating Generative AI into the context of the Fourth Industrial Revolution in developing countries, where technological change is a crucial determinant of progress and equitable growth.

Social

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Artificial Intelligence & Machine Learning
…but why now? It is not just ChatGPT.

Hiring for AI-related roles remains a challenge, though reported difficulty has decreased since 2022 for many roles.



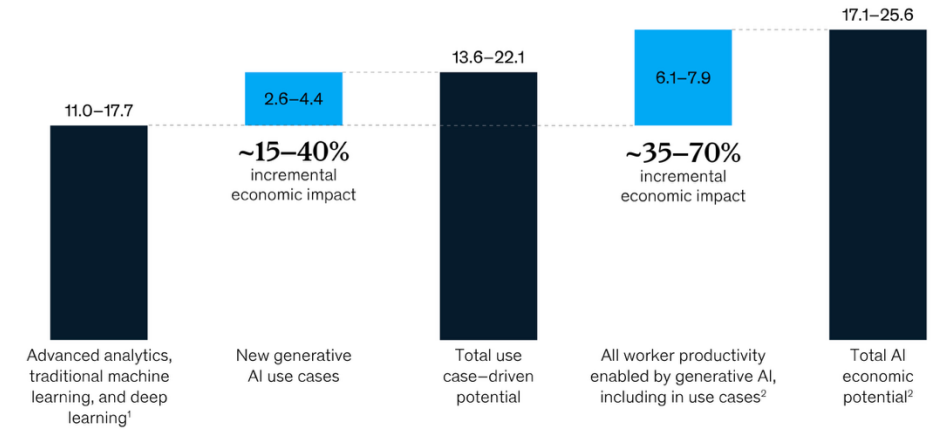
[1]Asked only of respondents whose organizations have adopted AI in at least 1 function and who said their organization hired the given role in the past 12 months. Respondents who said "easy," "neither difficult nor easy," or "don't know" are not shown.
[2]Not asked of respondents in 2022.
Source: McKinsey Global Survey on AI, 1,684 participants at all levels of the organization, April 11–21, 2023

McKinsey & Company

Report August 1, 2023
www.mckinsey.com/capabilities/quantumblack/our-insights/the-state-of-ai-in-2023-generative-ais-breakout-year

Generative AI could create additional value potential above what could be unlocked by other AI and analytics.

AI's potential impact on the global economy, $ trillion



[1]Updated use case estimates from "Notes from the AI frontier: Applications and value of deep learning," McKinsey Global Institute, April 17, 2018.
[2]The range of potential value from the combined impact of new generative AI use cases and the increased worker productivity they could enable is $6.1 trillion to $7.9 trillion, including revenue impacts conservatively translated into productivity impact as difference between total impact and cost-isolated impact.

McKinsey & Company

Report: August 25, 2023
https://www.mckinsey.com/featured-insights/mckinsey-explainers/whats-the-future-of-generative-ai-an-early-view-in-15-charts

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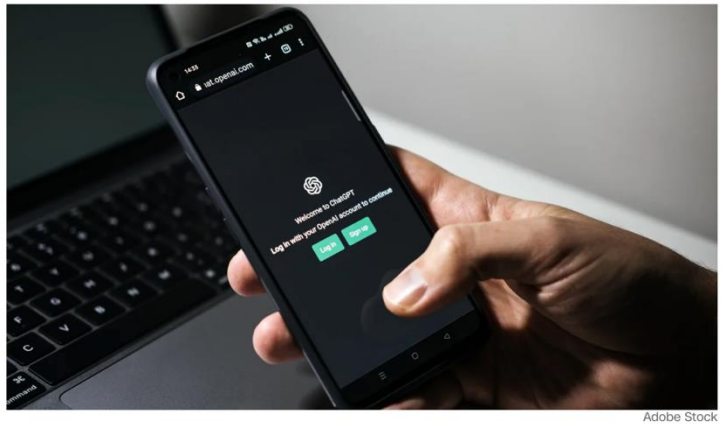
Artificial Intelligence & Machine Learning
What are the challenges in application?

Bias in Data

[1]

AI can be racist, sexist and creepy. What should we do about it?

Analysis by Zachary B. Wolf, CNN
Published 9:29 AM EDT, Sat March 18, 2023



Adobe Stock

The ChatGPT chatbot developed by OpenAI.

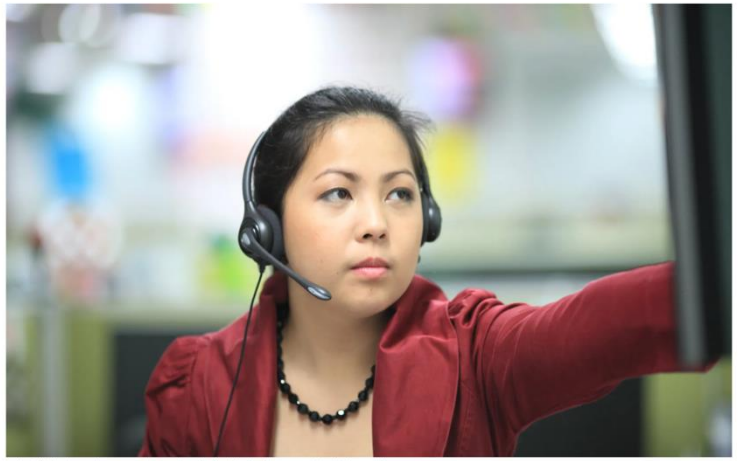
Labor Conditions

[2]

News • Policy & Finance

AI boom is dream and nightmare for workers in Global South

Lax labour regulations and low wages are the norm for data annotation workers in poorer nations, but many have no choice.



In the Philippines - long an outsourcing destination for its young, English-speaking population - freelance data annotation on platforms such as Upwork has become highly competitive. Image: ILO Asia-Pacific, CC BY-SA 3.0, via Flickr.

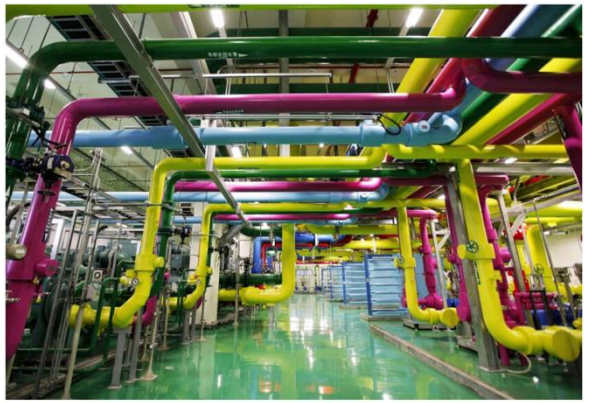
Thomson Reuters Foundation 6 minute read • March 16, 2023

Sustainability

[3]

Artificial Intelligence Is Booming—So Is Its Carbon Footprint

Greater transparency on emissions could also bring more scrutiny



Pipes pass through the chiller plant at a Google data center in Changhua, Taiwan. *Photographer: Ashley Pon/Bloomberg*

By Josh Saul and Dina Bass
9 March 2023 at 15:00 CET

https://edition.cnn.com/2023/03/18/politics/ai-chatgpt-racist-what-matters/index.html [1]
https://www.eco-business.com/news/ai-boom-is-dream-and-nightmare-for-workers-in-global-south/ [2]
https://www.bloomberg.com/news/articles/2023-03-09/how-much-energy-do-ai-and-chatgpt-use-no-one-knows-for-sure?embedded-checkout=true [3]

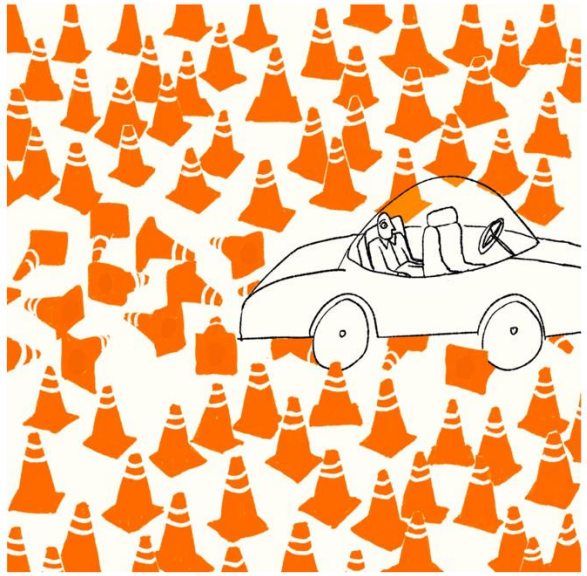
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Artificial Intelligence & Machine Learning
Is it safe though? Appropriate Reliance.

OPINION
GUEST ESSAY

Autonomous Vehicles Are Driving Blind

Oct. 11, 2023



Jared Nangle

[1]

ChatGPT provided better customer service than his staff. He fired them.

Artificial intelligence is rapidly changing the world of customer service and call centers. Developing economies worry they'll face the brunt.

By Pranshu Verma

October 3, 2023 at 7:00 a.m. EDT



(Davide Bonazzi for The Washington Post)

[2]

Appropriate Reliance on AI Advice: Conceptualization and the Effect of Explanations

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ABSTRACT

AI advice is becoming increasingly popular, e.g., in investment and medical treatment decisions. As this advice is typically imperfect, decision-makers have to exert discretion as to whether actually follow that advice: they have to "appropriately" rely on correct and turn down incorrect advice. However, current research on appropriate reliance still lacks a common definition as well as an operational measurement concept. Additionally, no in-depth behavioral experiments have been conducted that help understand the factors influencing this behavior. In this paper, we propose Appropriateness of Reliance (AoR) as an underlying, quantifiable two-dimensional measurement concept. We develop a research model that analyzes the effect of providing explanations for AI advice. In an experiment with 200 participants, we demonstrate how these explanations influence the AoR, and, thus, the effectiveness of AI advice. Our work contributes fundamental concepts for the analysis of reliance behavior and the purposeful design of AI advisors.

CCS CONCEPTS

• **Human-centered computing** → **Empirical studies in HCI**; • **Computing methodologies** → **Artificial intelligence**.

KEYWORDS

Appropriate Reliance, Explainable AI, Human-AI Collaboration, Human-AI Complementarity

ACM Reference Format:
Max Schemmer, Niklas Kühl, Carina Benz, Andrea Bartos, and Gerhard Satzger. 2023. Appropriate Reliance on AI Advice: Conceptualization and the Effect of Explanations. In *28th International Conference on Intelligent User Interfaces (IUI '23), March 27–31, 2023, Sydney, NSW, Australia.* ACM, New York, NY, USA, 13 pages. https://doi.org/10.1145/3581641.3584066

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IUI '23, March 27–31, 2023, Sydney, NSW, Australia

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ACM ISBN 979-8-4007-0106-1/23/03…$15.00
https://doi.org/10.1145/3581641.3584066

1 INTRODUCTION

Most important decisions are made by calling upon advisors. While in the past advice was typically obtained from human experts, nowadays advisors based on artificial intelligence (AI) are becoming increasingly frequent in research and practice [33, 34]. For example, AI advises medical professionals in breast cancer screening [48], or in loan decisions[16].

In the past, research has predominantly focused on maximizing the reliance [35], trust [66], utilization [2], compliance [37], or acceptance [65] of AI advice [34]. With certain nuances, all these terms basically describe a concept that aims at maximizing the amount of AI advice that a human decision-maker eventually follows, i.e. maximizing the AI reliance. Recently, a new line of research has emerged that argues that maximising AI reliance does not fully exploit the potential of state-of-the-art human-AI decision making [6, 9, 62, 78]. We summarize the reasons for this line of thought in three main categories:

Increasing usage of imperfect AI advisors. First, prior research on AI advice often assumed "perfect" advice [34]. This makes sense if one considers narrow application spaces, e.g. performing deterministic algebra. But AI nowadays is used for more complex tasks [19] which increases both the number and severity of AI errors. Therefore, generally accepting AI advice would also include the acceptance of incorrect AI advice: Assume that without an AI advisor's intervention, a physician would have diagnosed cancer, yet in the setting with AI advice had been misled by an incorrect AI advice and, thus, had failed to detect the cancer.

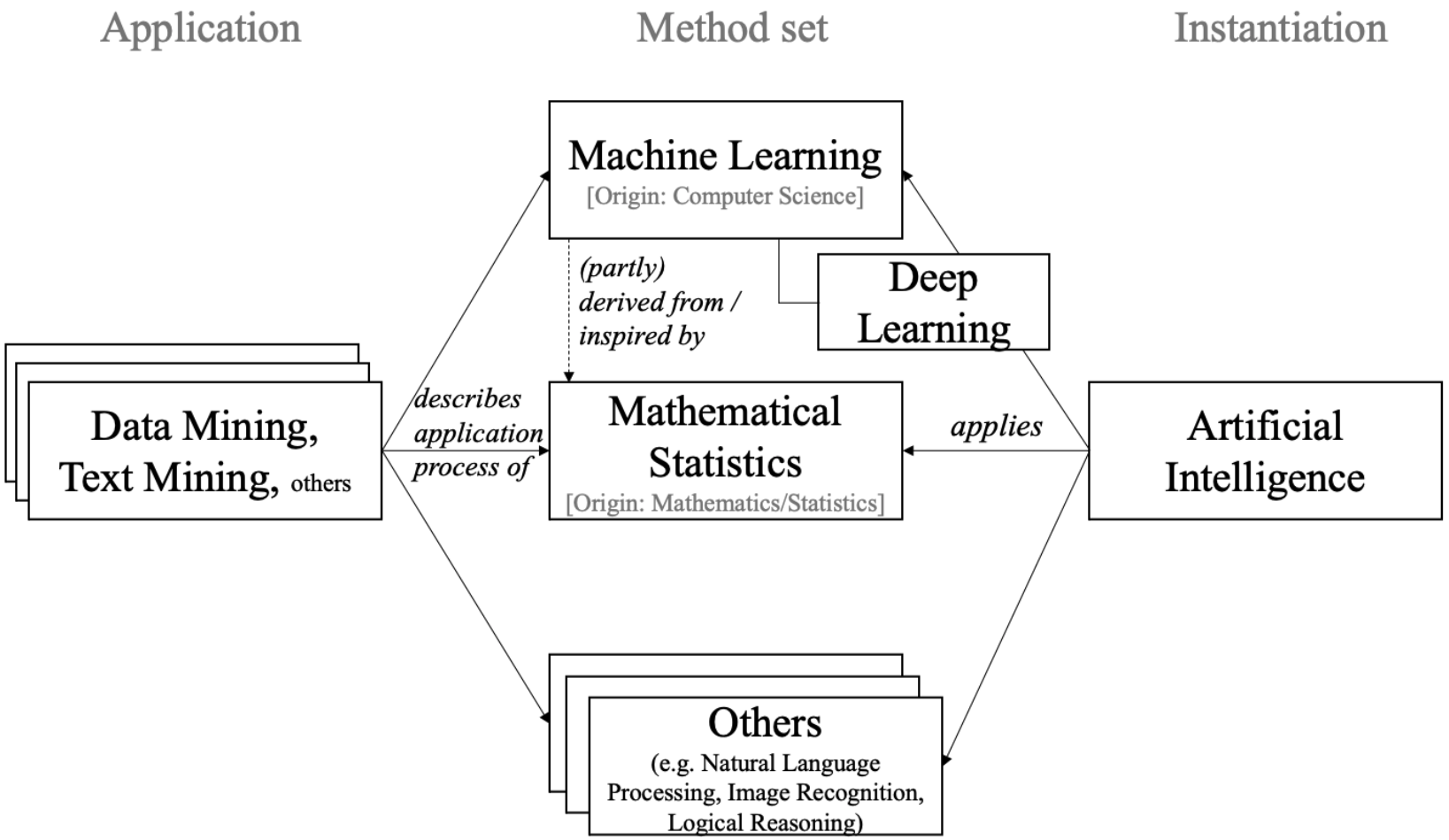
Increasing alignment of the objectives of human decision-makers and AI advisors. Second, in previous research, the goal of human or AI advisors was often inconsistent with the goal of the decision-maker, e.g., financial advisors wanted to persuade a client to purchase certain financial products that would yield the highest financial benefit for their own bonus. In this situation, AI designers do not want customers to differentiate good from bad advice but simply increase acceptance of advice. Today, however, AI advisors are often specifically designed to enhance human decision-making [41]. The advice seekers can design the advisor based on individual goals and with desired features such as "honest" explainability. This honesty might be missing if the inherent goal of the advisor is different from the advice seeker's. If the AI is not designed in alignment with human goals, it is often not in the AI designer's

[3]

https://www.nytimes.com/2023/10/11/opinion/driverless-cars-san-francisco.html#:~:text=When%20it%20comes%20to%20self,drive%20thousands%20of%20cars%20through [1]
https://www.washingtonpost.com/technology/2023/10/03/ai-customer-service-jobs/ [2]
https://dl.acm.org/doi/abs/10.1145/3581641.3584066 [3]

Prof. Dr. Niklas Kühl

Artificial Intelligence & Machine Learning
So many terms – What do they mean?



Prof. Dr. Niklas Kühl

Artificial Intelligence & Machine Learning
What is AI? Is Mark Zuckerberg really talking about AI?



[1]

In Mark Zuckerberg's US senate hearing in April 2018, he stressed the necessary capabilities of Facebook's

"**AI tools** (...) to (...) identify hate speech (...)" or "(...) terrorist propaganda"

The Washington Post (2018), Transcript of Mark Zuckerberg's Senate hearing [1]

Prof. Dr. Niklas Kühl

Artificial Intelligence & Machine Learning
Different views on AI definitions.

Research Currents:
Different scientific viewpoints on what constitutes important aspects of AI research

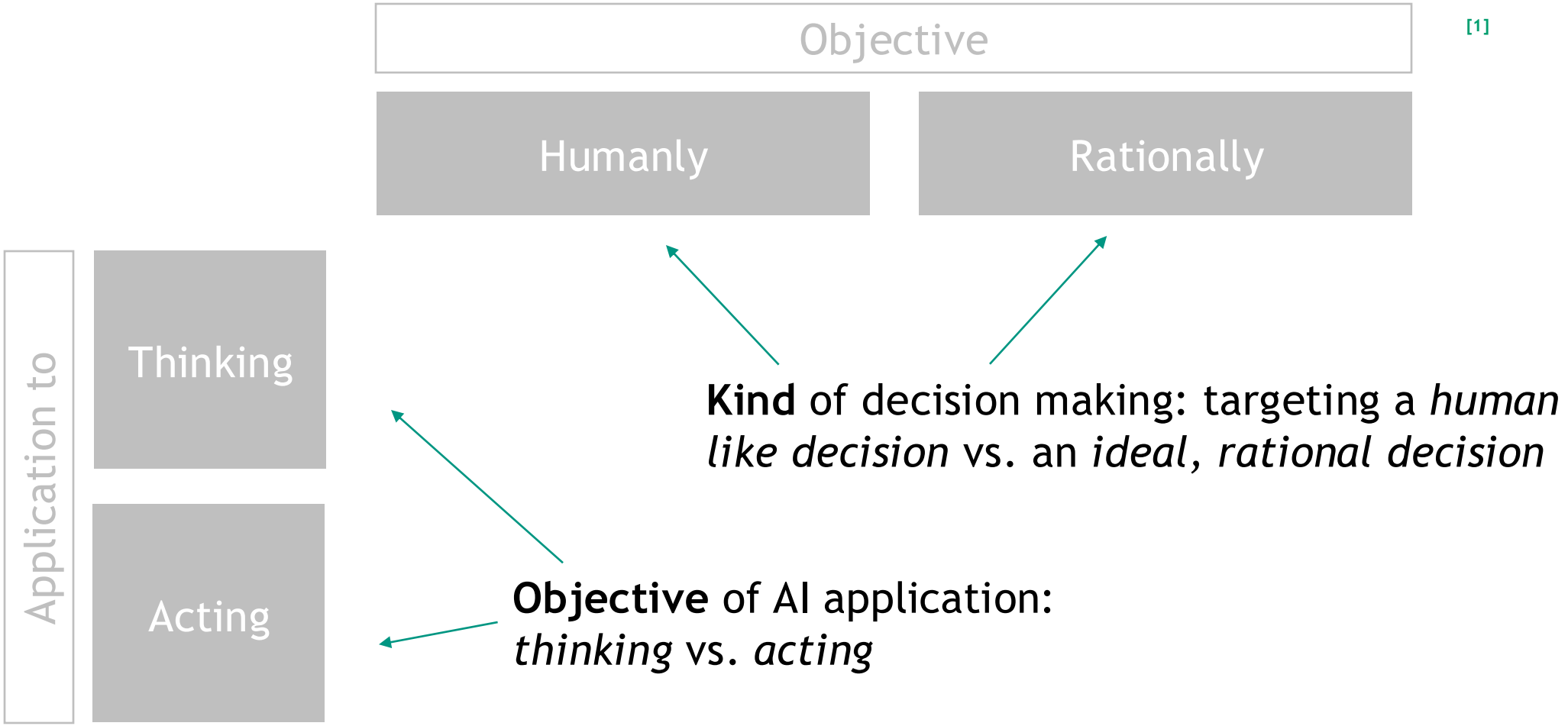
Agent-based AI:
Agents are the central instantiation of AI



Level of Intelligence:
AI can be differentiated by its cognitive capabilities

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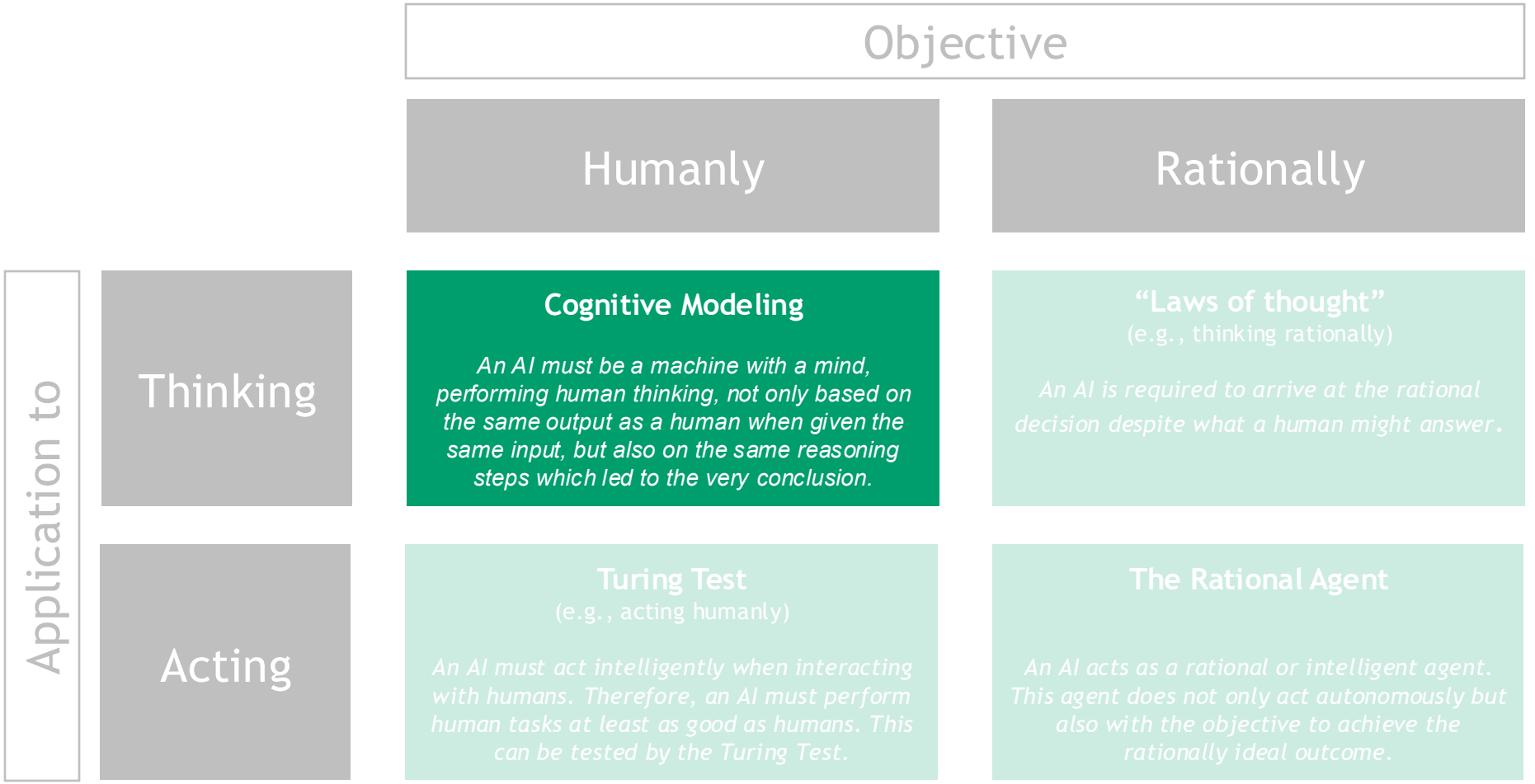
Artificial Intelligence & Machine Learning
Research currents of AI.



Russel and Norvig (2015), Artificial Intelligence: A Modern Approach ; Haugeland (1989), Artificial Intelligence: The very idea ; Bellmann (1978), An Introduction to Artificial Intelligence: Can Computers Think? ; Newell and Simon (1961), GPS, a program that simulates human thought ; Kühl, Goutier, Hirt, Satzger (2019): Machine Learning in Artificial Intelligence: Towards a Common Understanding [1]

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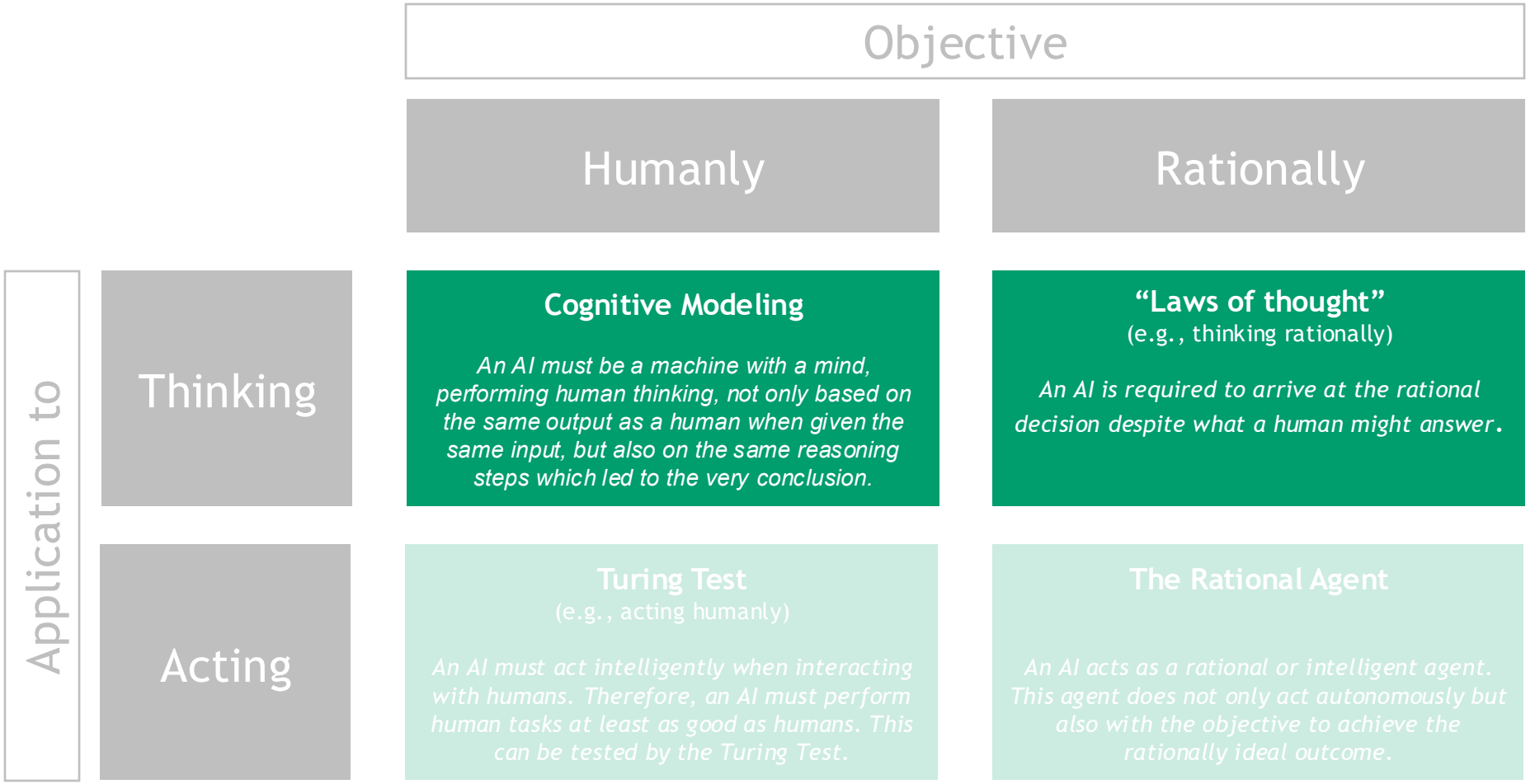
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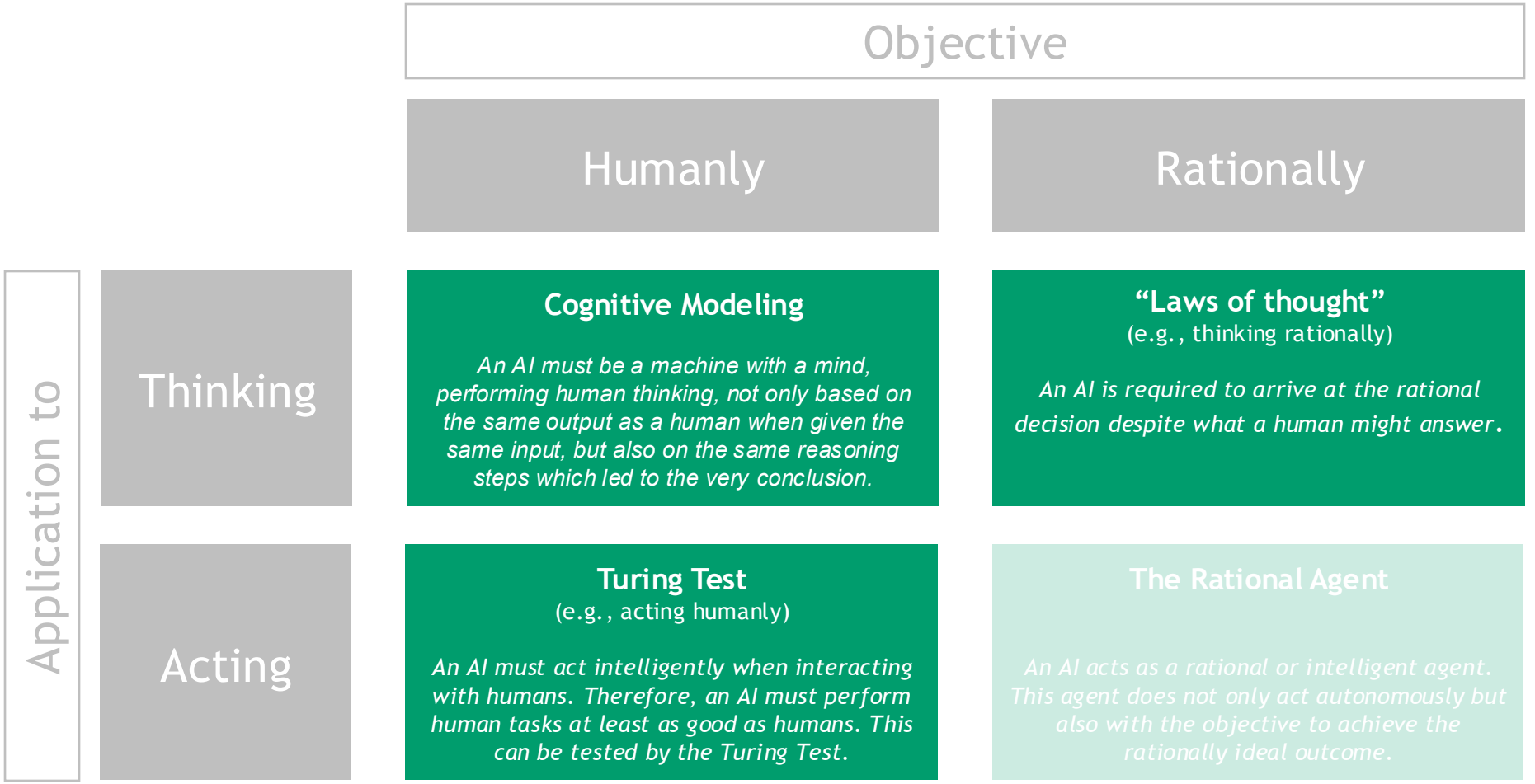
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Artificial Intelligence & Machine Learning
Research currents of AI.

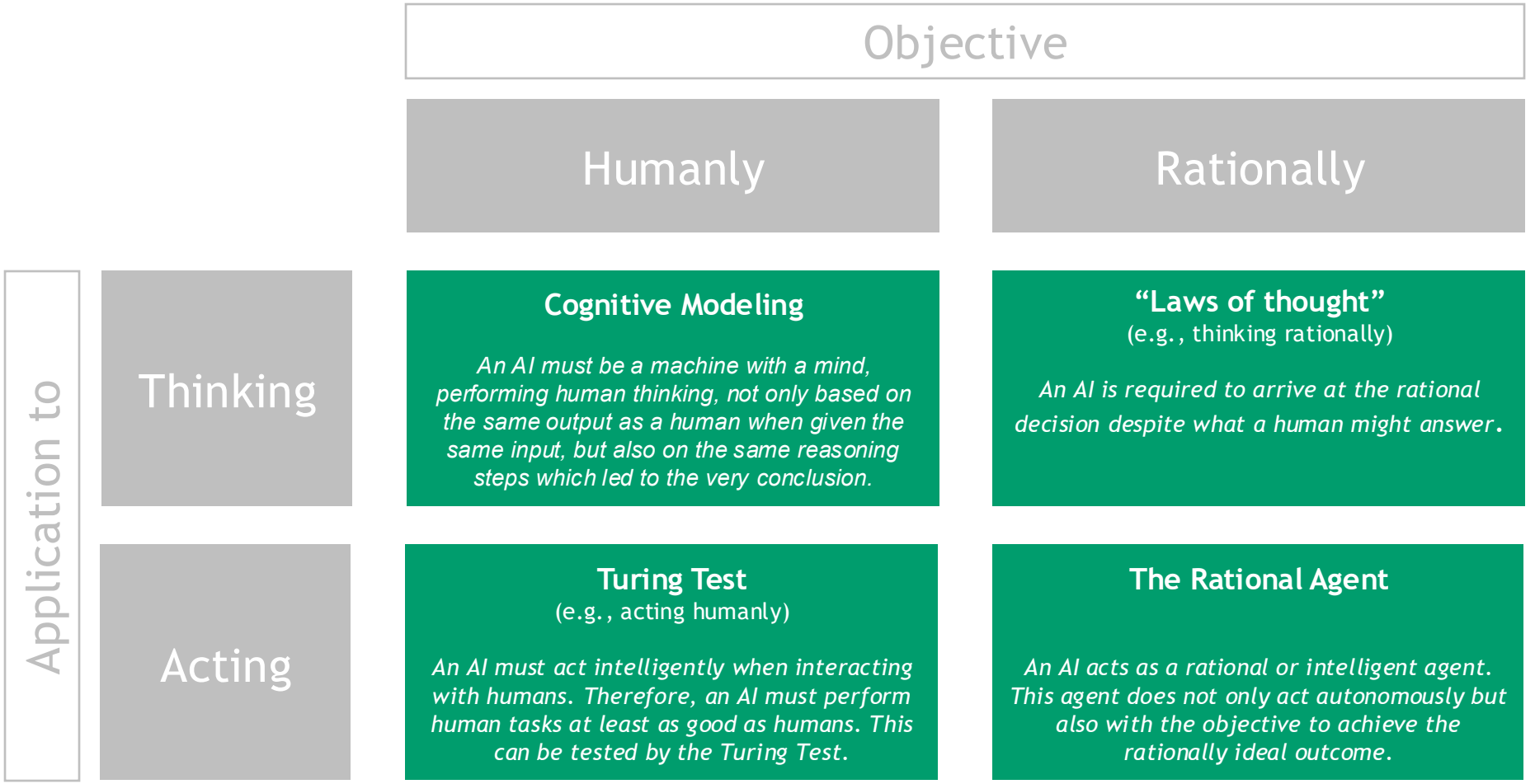


[1]

Russel and Norvig (2015), Artificial Intelligence: A Modern Approach ; Haugeland (1989), Artificial Intelligence: The very idea ; Bellmann (1978), An Introduction to Artificial Intelligence: Can Computers Think? ; Newell and Simon (1961), GPS, a program that simulates human thought ; Kühl, Goutier, Hirt, Satzger (2019): Machine Learning in Artificial Intelligence: Towards a Common Understanding [1]

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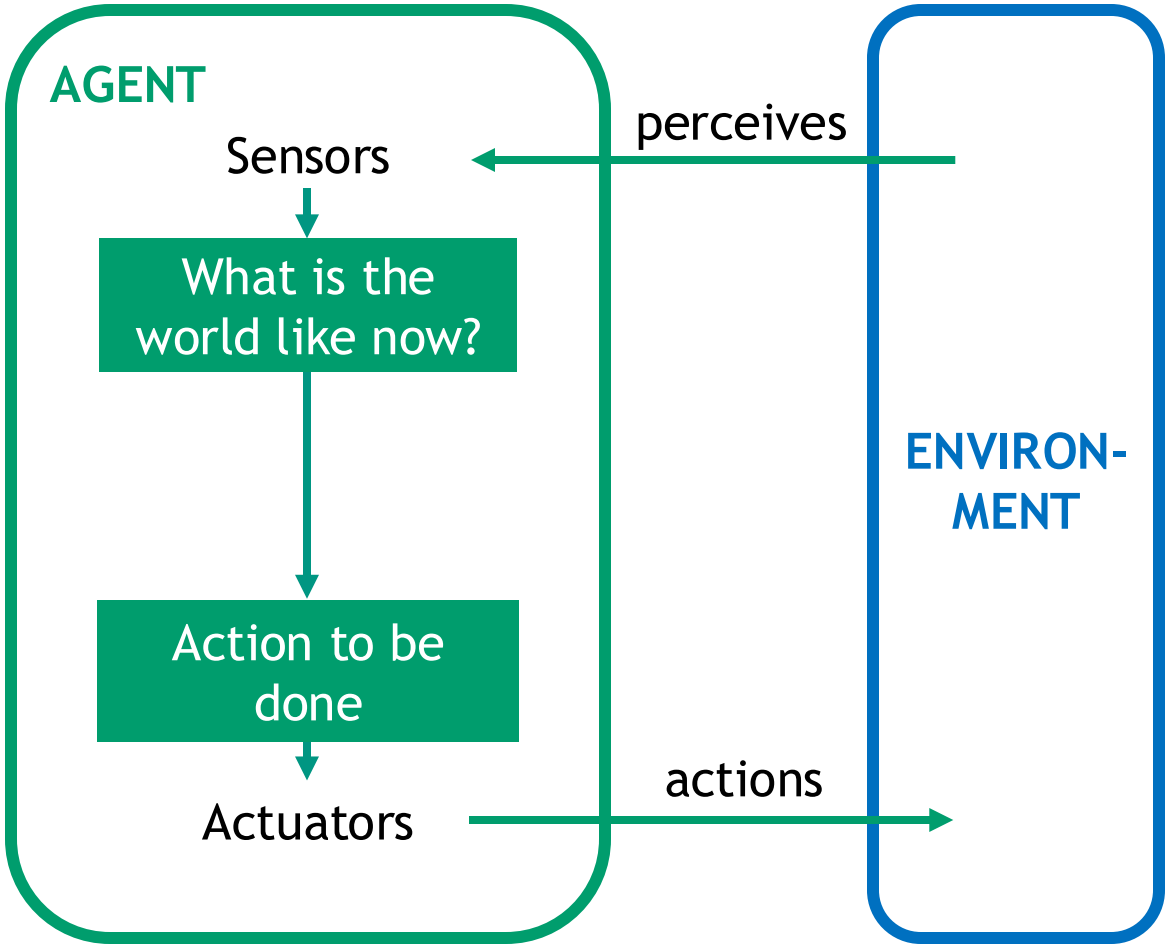
Artificial Intelligence & Machine Learning
Research currents of AI.



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Artificial Intelligence & Machine Learning
Definition of AI: An agent-oriented view.



[1]

- An intelligent agent interacts with an environment
- The agent perceives states of an environment and decides about what actions to take depending on these states
- Receiving sensors and executing actuators exist in a circular relationship

Russel and Norvig (2015), Artificial Intelligence: A Modern Approach [1]

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Artificial Intelligence & Machine Learning
The PEAS model.

Example:
The agent is a taxi driver



[1]

Performance Measures	Environment
Safe, fast, legal, comfortable trips, maximize profit, …	Roads, other traffic, pedestrians, customers, …

Actuators	Sensors
Steering, accelerator, brake, signal, horn, display, …	Cameras, sonar, speed-ometer, GPS, accelerometer, engine sensors, keyboard, …

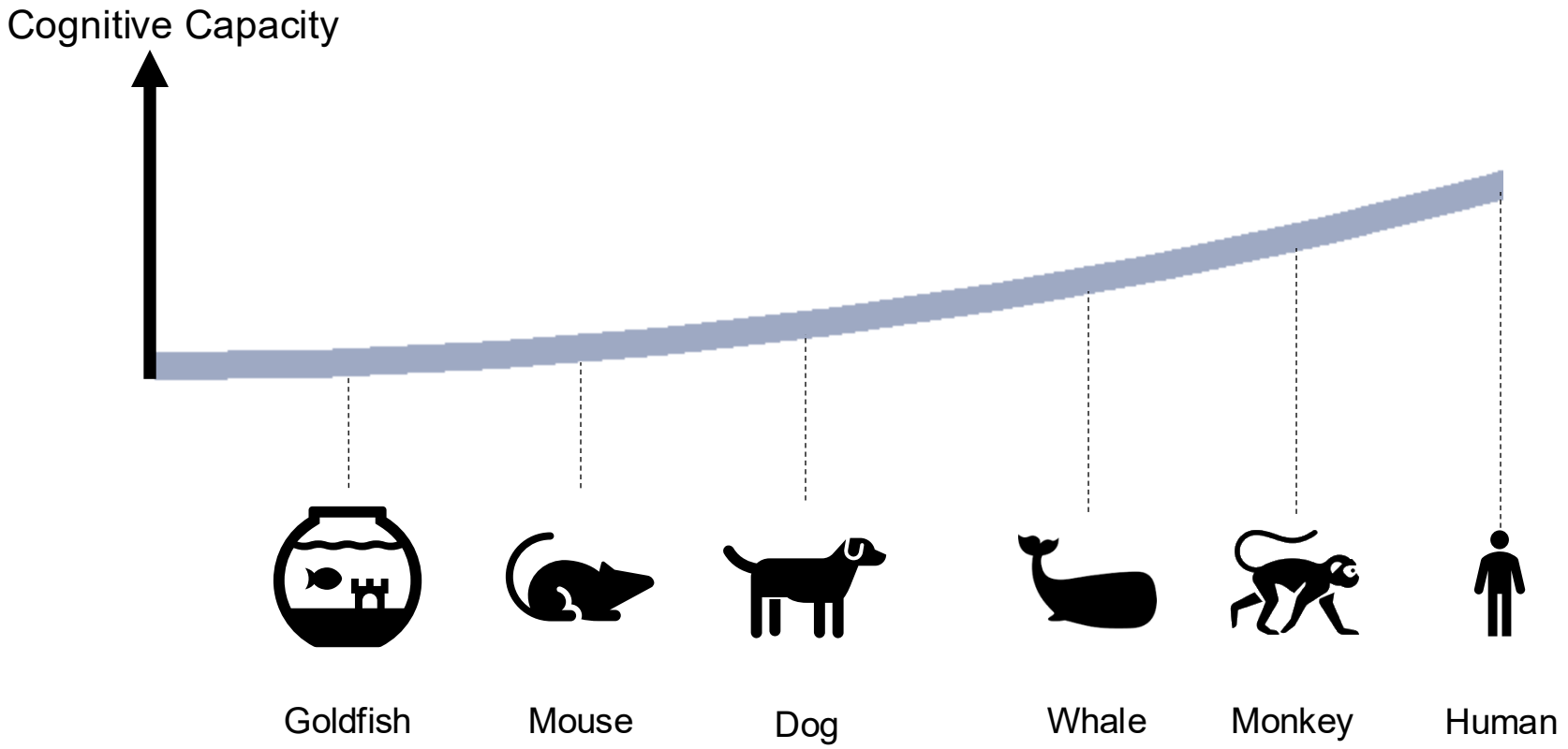
A simple aggregation of an agent's performance measures, in an environment with actuators and sensors (PEAS)

Russel and Norvig (2015), Artificial Intelligence: A Modern Approach [1]

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Artificial Intelligence & Machine Learning
Definition of AI: An intelligence-level-oriented view

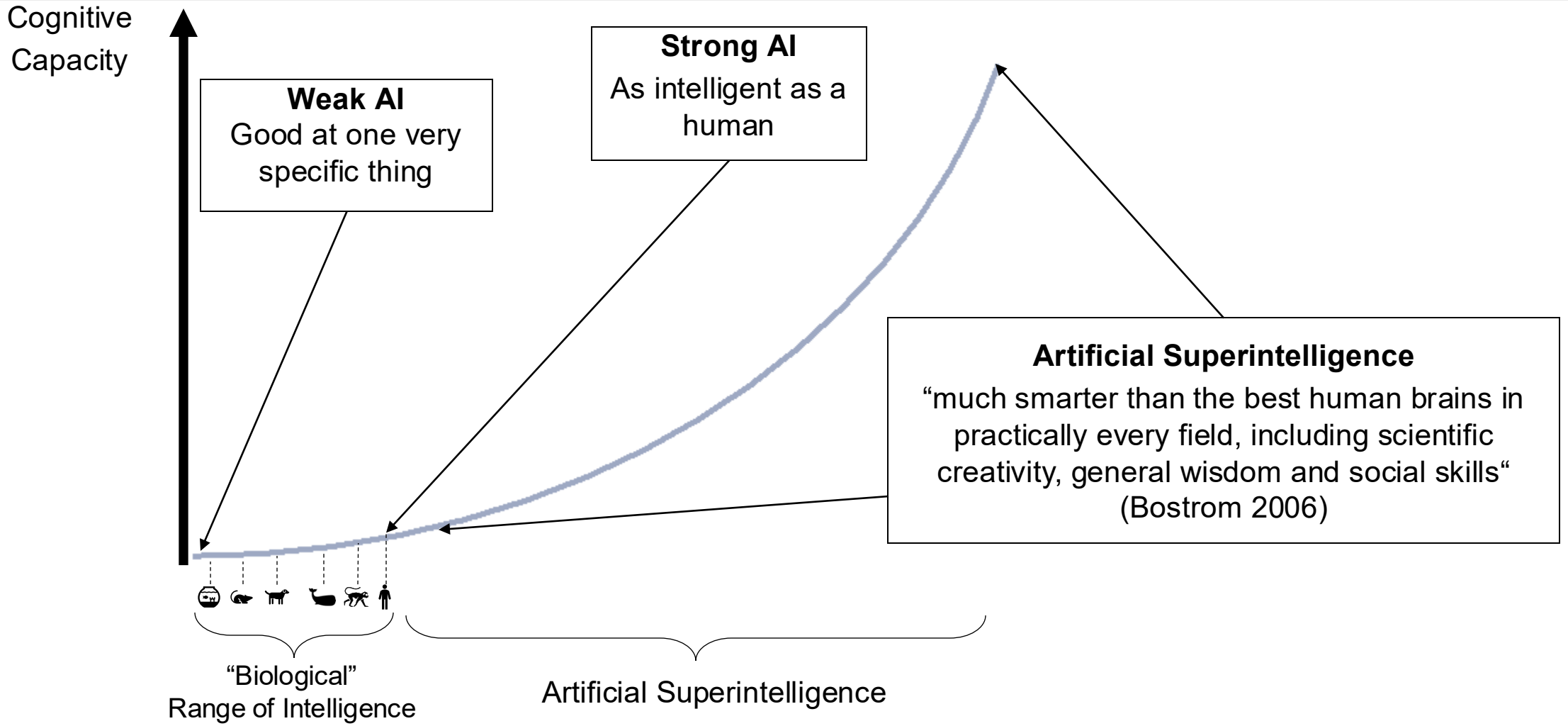
Intelligence: *The ability to perceive information, to retain it as knowledge and to apply it towards adaptive behaviors within an environment.* [1]



Cairo (2011) [1]

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Artificial Intelligence & Machine Learning
What “levels” of AI can we define?



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Artificial Intelligence & Machine Learning
Three different types of machine learning are common.

Machine Learning Types

Supervised Learning

- Learn the mapping from the input to the output
- Typical: regression / classification
- Simplified example: Let a child sort toy cars and tell it there are sports cars and SUVs with distinct properties



[1]

Unsupervised Learning

- Identify previously unknown patterns in data
- Typical: clustering / association rules
- Simplified example: Let a child sort toy cars and let the child determine how to arrange/cluster them



[2]

Reinforcement Learning

- Get feedback (rewards) on actions taken
- Simplified example: Child picks toy car and labels it (“action”) as truck vs. passenger car and gets response (”reward”) from his dad

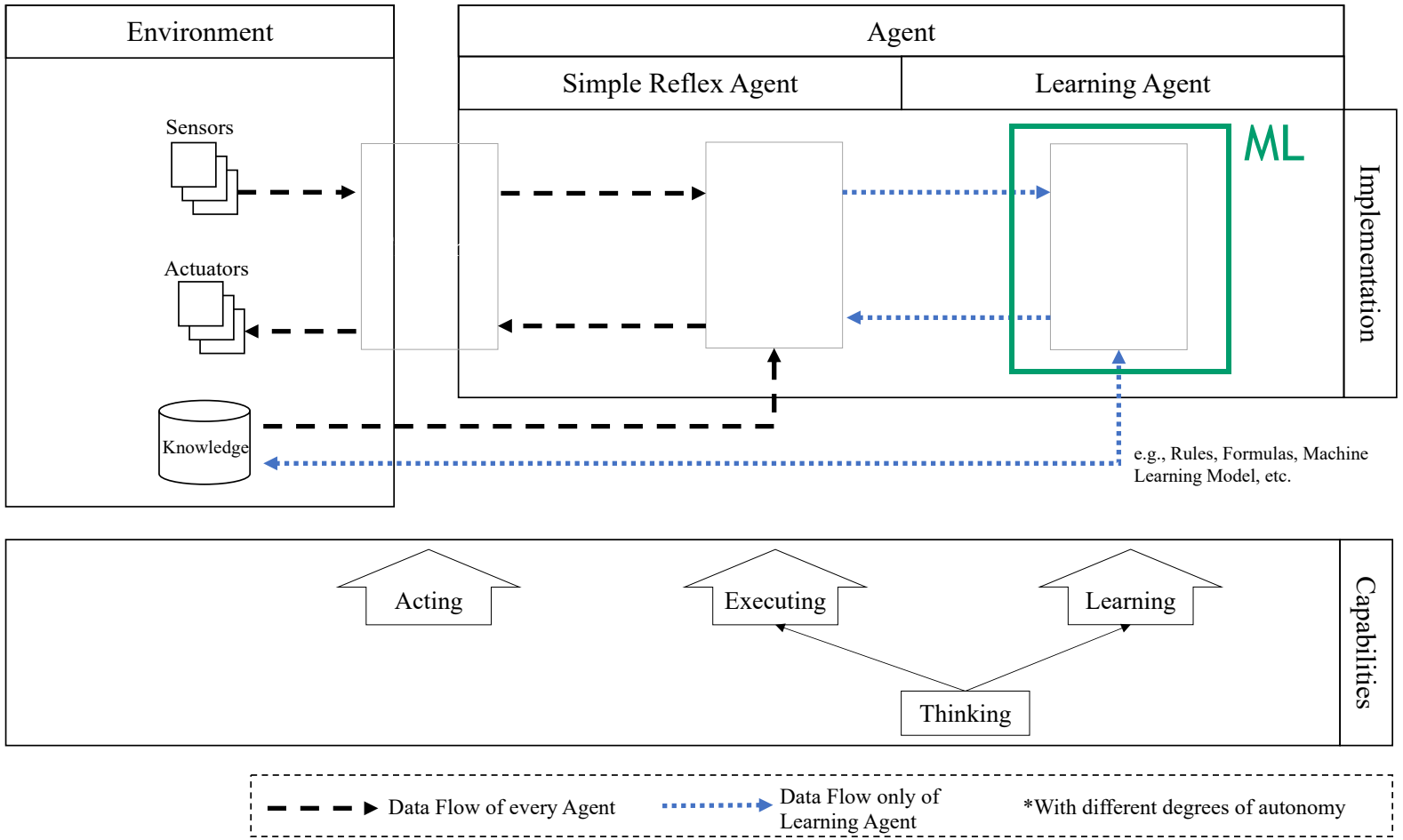


[3]

https://www.pexels.com/photo/young-girls-playing-with-toy-cars-4491563/ [1]
https://www.pickpik.com/child-toys-on-the-floor-floor-fun-car-82464 [2]
https://www.pexels.com/photo/close-of-father-and-son-playing-with-plastic-toys-outdoors-4933608/ [3]

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What role does ML play in AI?



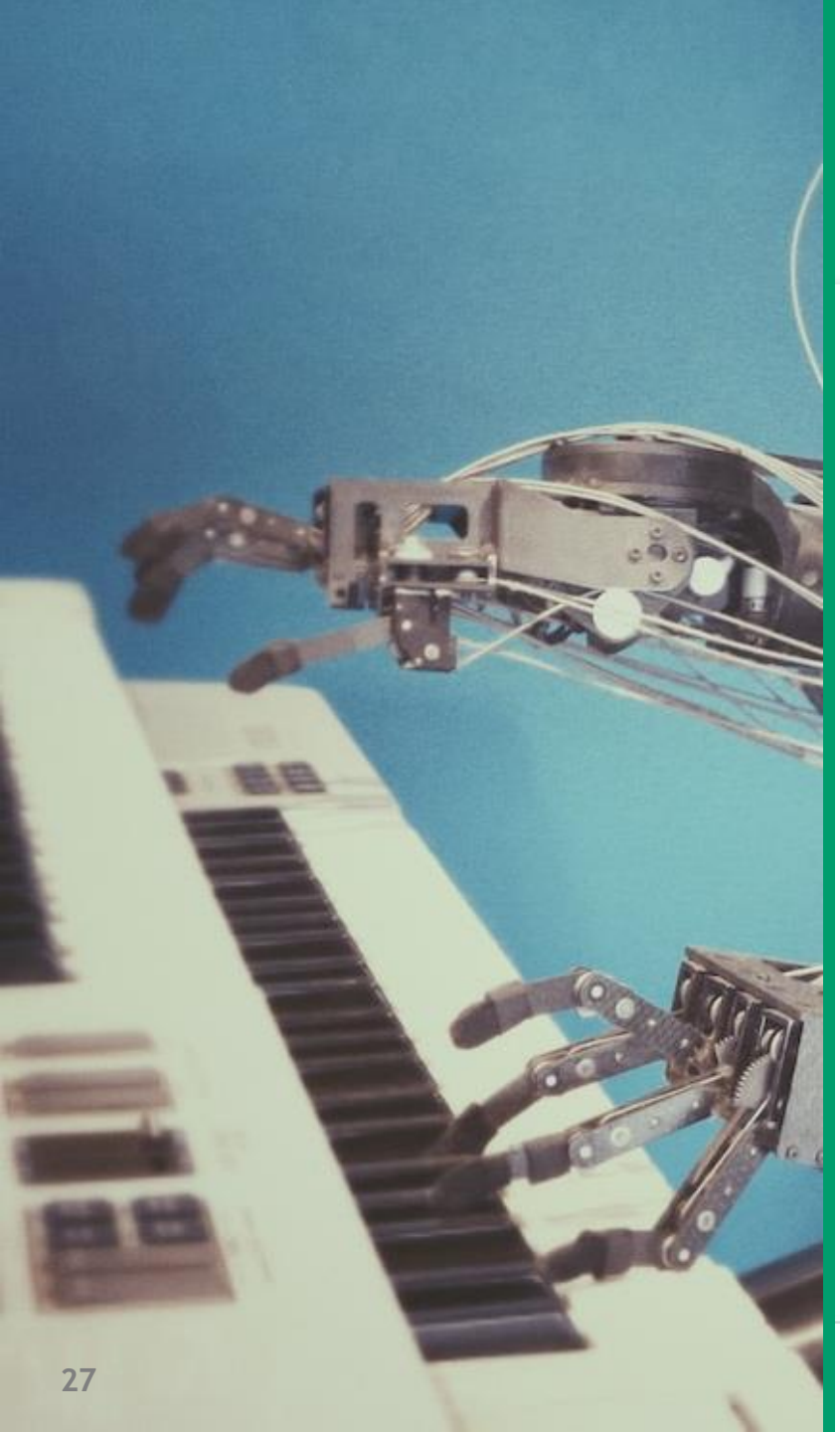
[1]

Kühl, Schemmer, Goutier, Satzger (2022): Artificial intelligence and machine learning [1]

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Artificial Intelligence & Machine Learning
So... was Mark Zuckerberg right?

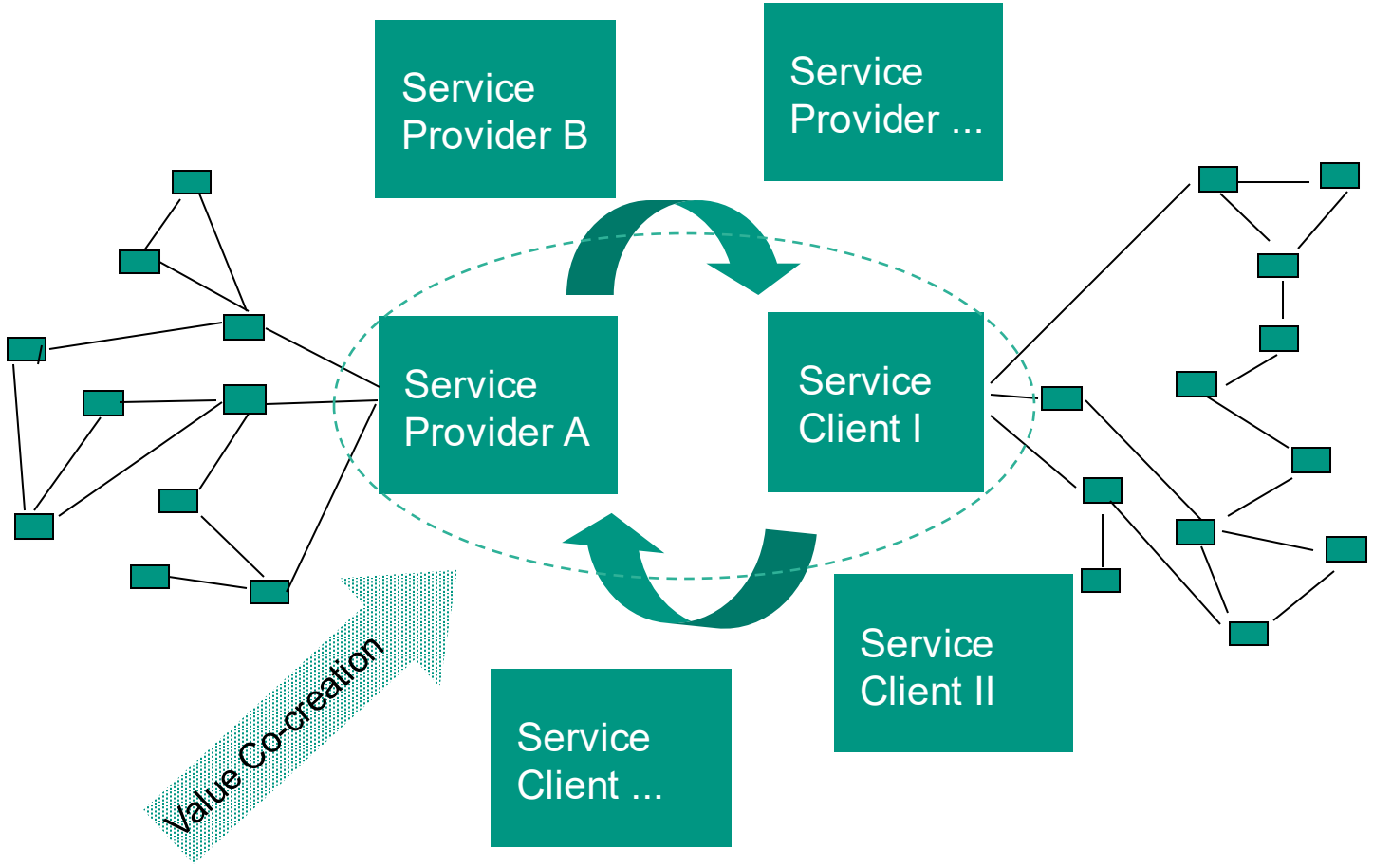
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1. Organizational
2. Artificial Intelligence & Machine Learning
3. **Systems Thinking**
4. The AI Lifecycle

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Service Systems
What are (Service) Systems?



Service System comprises ***"service providers and service clients working together to coproduce value in complex value chains or networks"*** [1]

Spohrer et al. [1]

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Service Systems
Examples



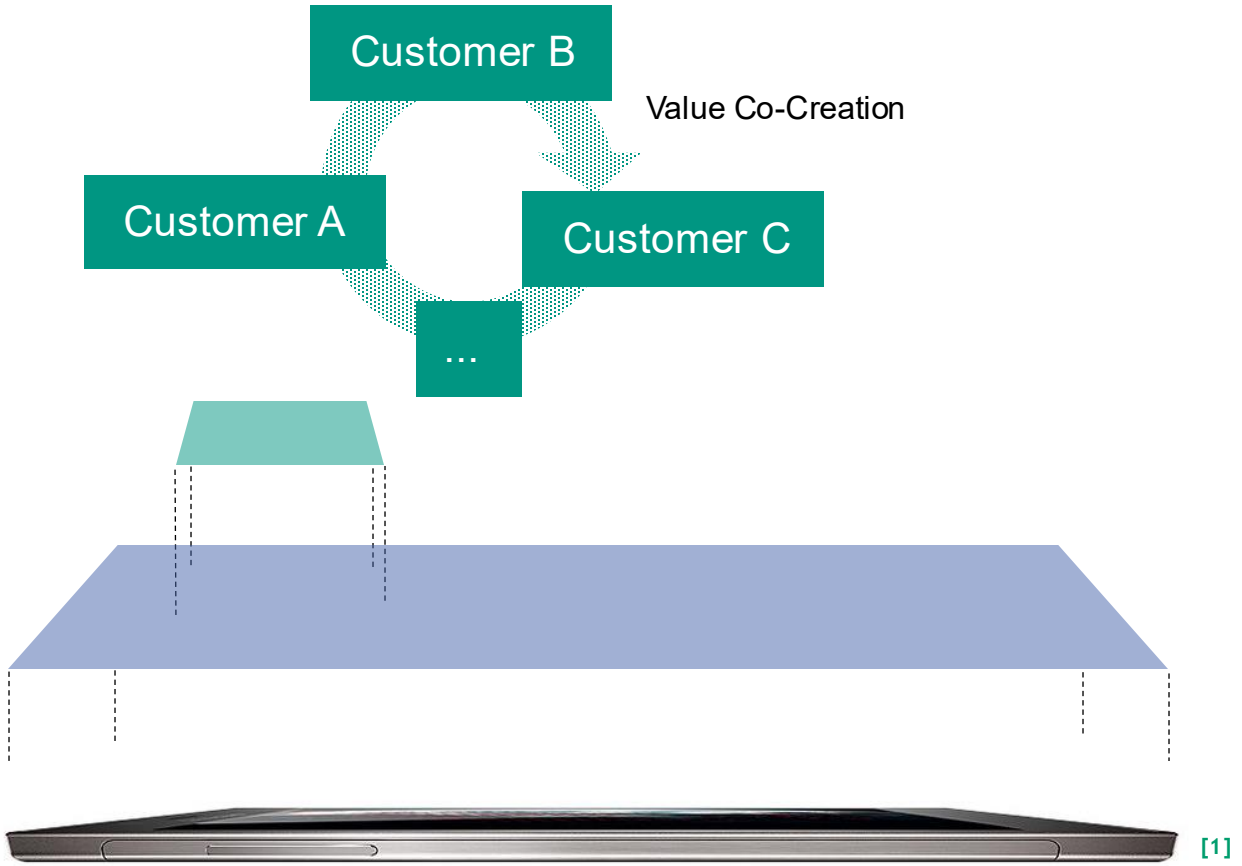
[1]

[2]

Maglio, P.P., Vargo, S.L., Caswell, N. et al. Inf Syst E-Bus Manage (2009) 7: 395. https://doi.org/10.1007/s10257-008-0105-1 [1]
https://freerangestock.com/photos/94025/person-washing-car-with-sponge.html [2]

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Service Systems
Examples



[1]

Application Provider, e.g., TikTok

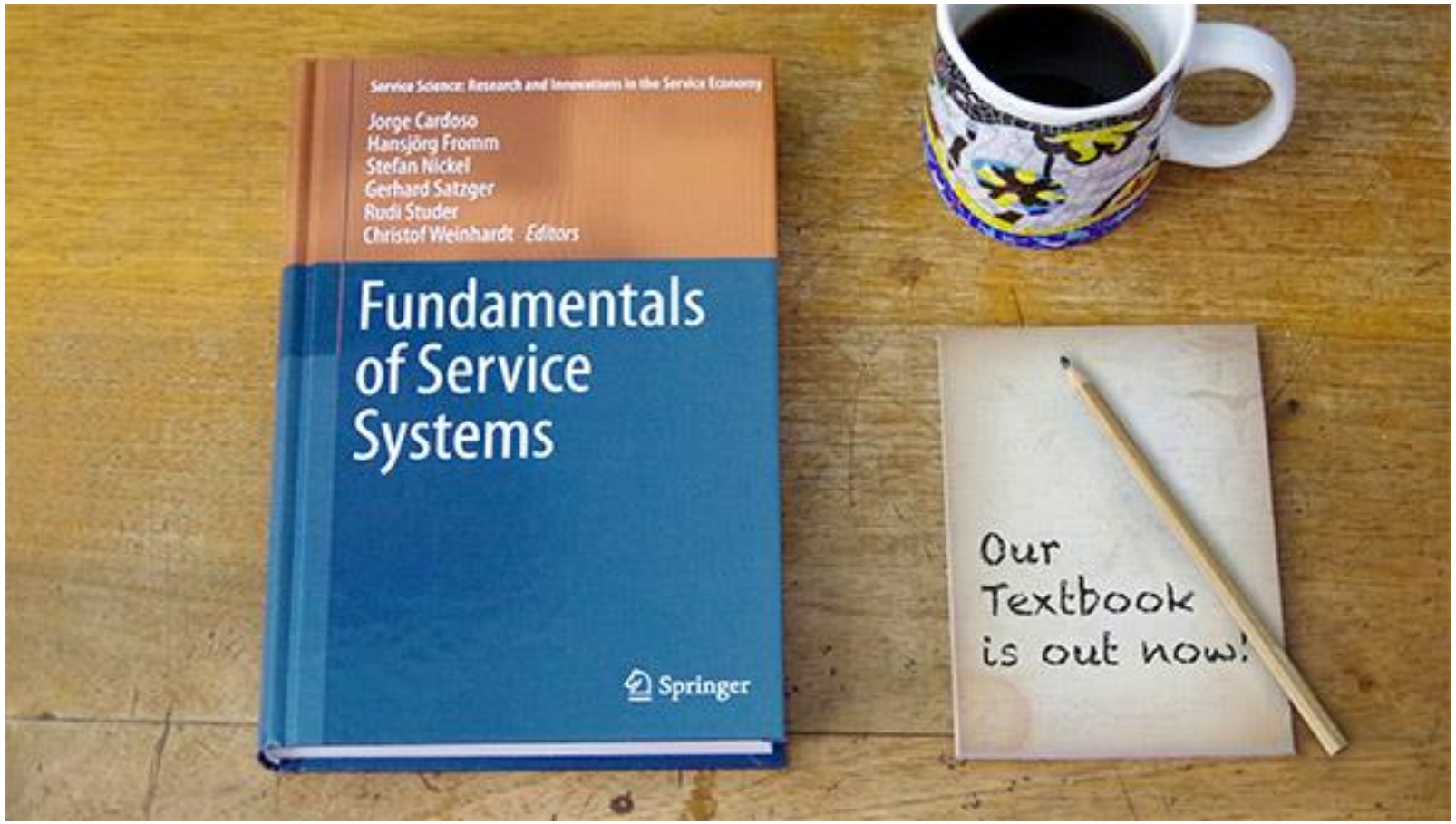
Operation System Provider, e.g., Google

Smartphone Provider, e.g., Samsung

https://www.flickr.com/photos/blltz/9369660668/ [1]

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Service Systems
The first textbook of Service Systems is available



[1,2]

Fundamentals of Service Systems. (2015). Deutschland: Springer International Publishing. [1]
https://fundamentals-of-service-systems.ksri.kit.edu/ [2]

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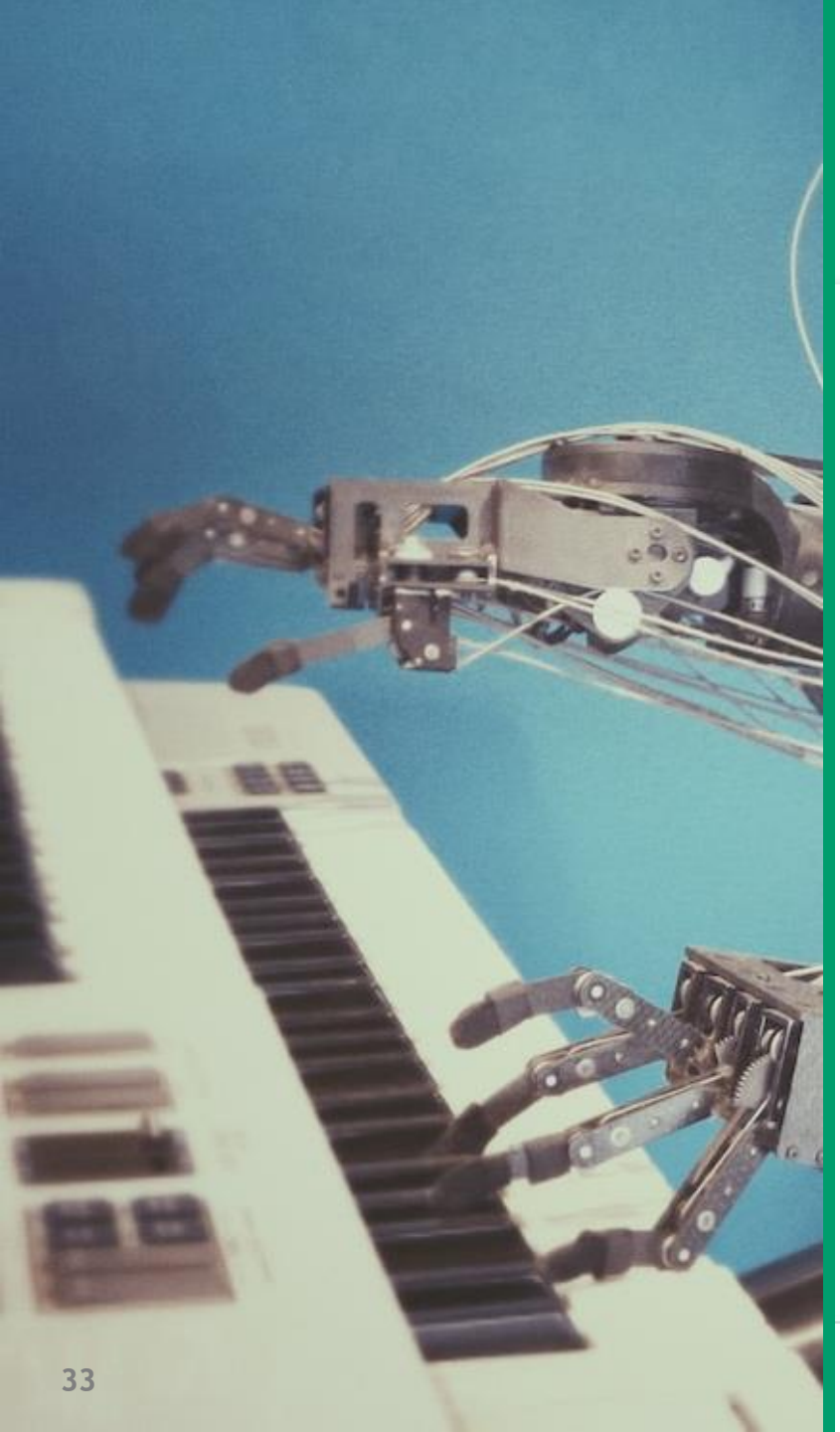
Service Systems
…as well as one on their management



[1]

Smart Service Management: Design Guidelines and Best Practices. (2021). Deutschland: Springer International Publishing.[1]

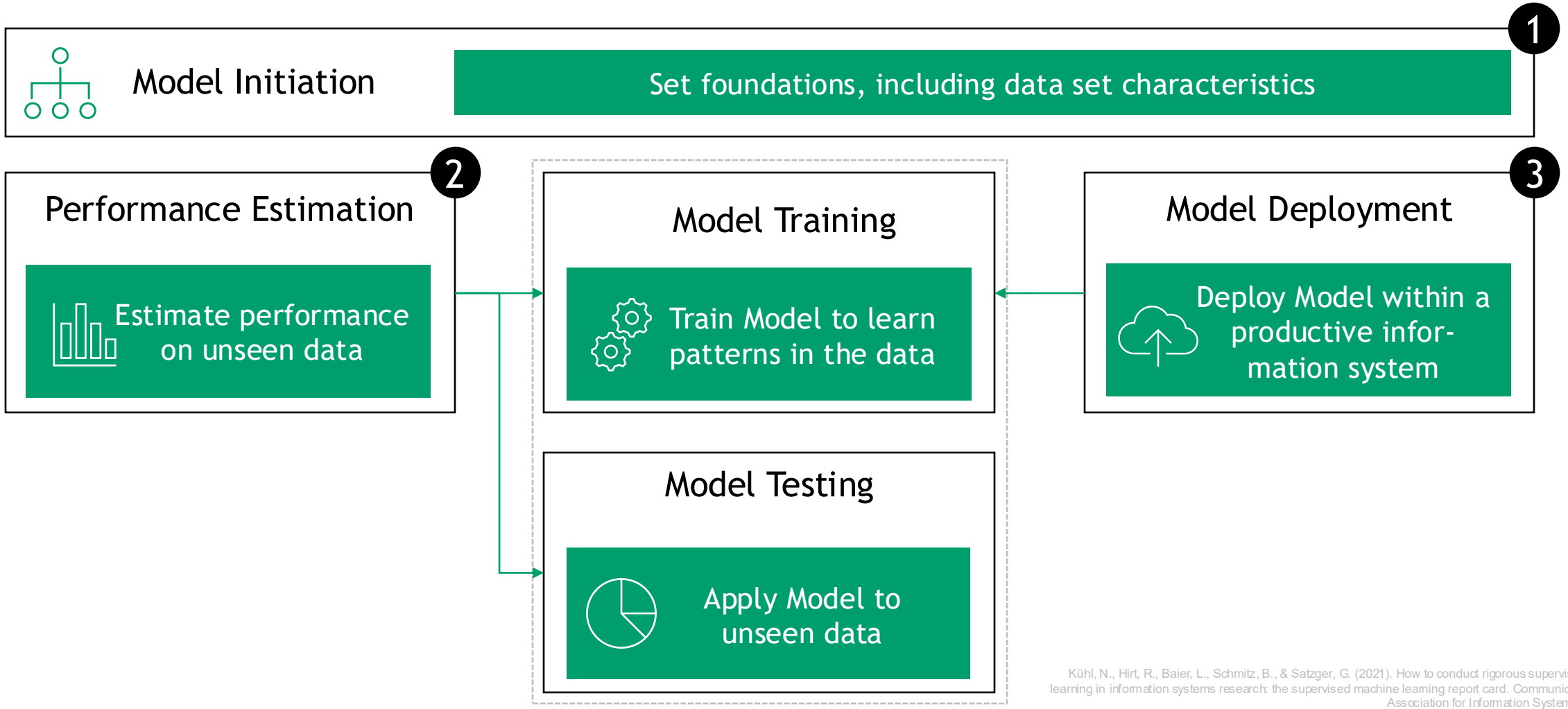
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1. Organizational
2. Artificial Intelligence & Machine Learning
3. Systems Thinking
4. **The AI Lifecycle**

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The AI Lifecycle
Required steps for supervised machine learning

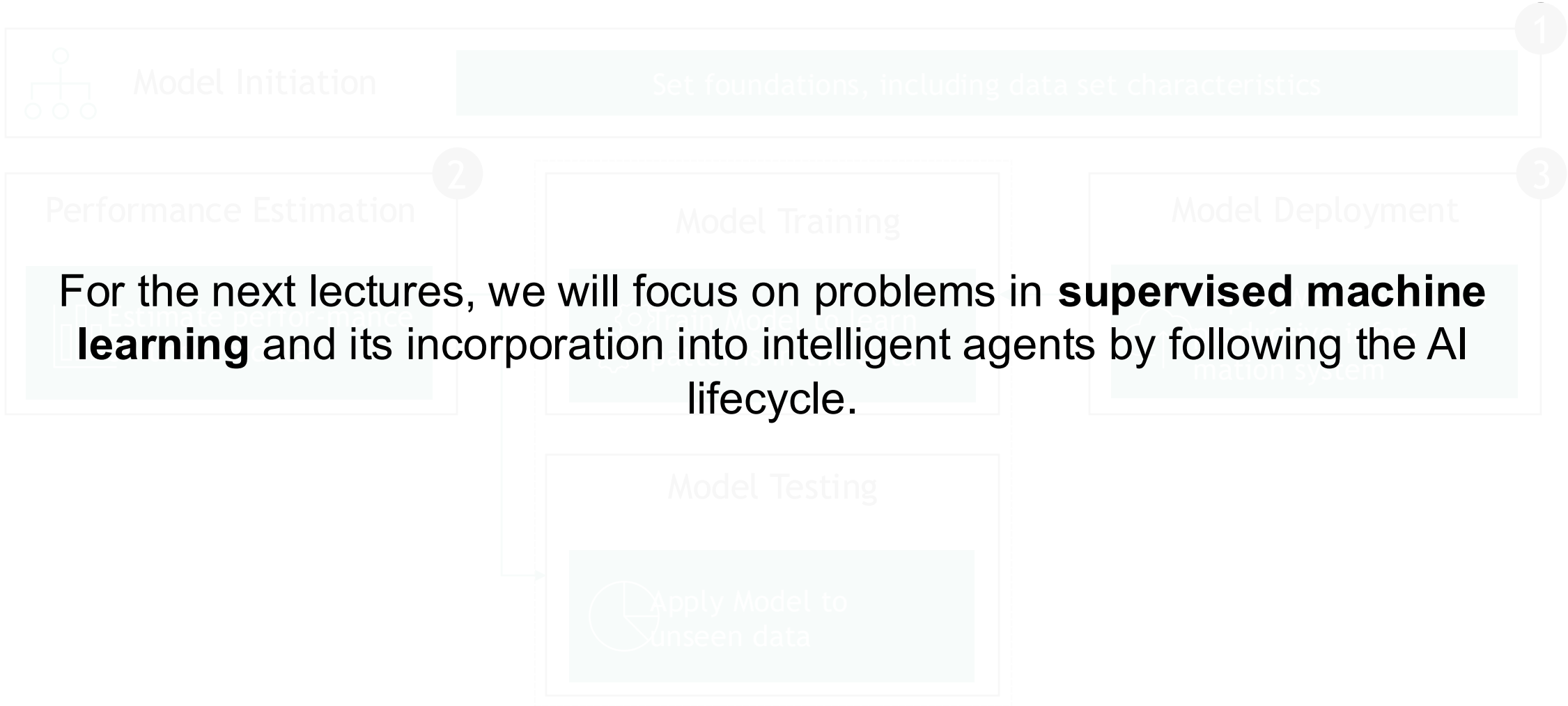


Kühl, N., Hirt, R., Baier, L., Schmitz, B., & Satzger, G. (2021). How to conduct rigorous supervised machine learning in information systems research: the supervised machine learning report card. Communications of the Association for Information Systems, 48(1), 46.

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The AI Lifecycle

Required steps for supervised machine learning.



For the next lectures, we will focus on problems in **supervised machine learning** and its incorporation into intelligent agents by following the AI lifecycle.

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